

### Question #1 of 139

Question ID: 1479918

During the course of a multiple regression analysis, an analyst has observed several items that she believes may render incorrect conclusions. For example, the coefficient standard errors are too small, although the estimated coefficients are accurate. She believes that these small standard error terms will result in the computed  $t$ -statistics being too big, resulting in too many Type I errors. The analyst has *most likely* observed which of the following assumption violations in her regression analysis?

A) Positive serial correlation.



B) Homoskedasticity.



C) Multicollinearity.



#### Explanation

Positive serial correlation is the condition where a positive regression error in one time period increases the likelihood of having a positive regression error in the next time period. The residual terms are correlated with one another, leading to coefficient error terms that are too small.

(Module 1.3, LOS 1.i)

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### Question #2 of 139

Question ID: 1472066

When two or more of the independent variables in a multiple regression are correlated with each other, the condition is called:

A) multicollinearity.



B) conditional heteroskedasticity.



C) serial correlation.



#### Explanation

*Multicollinearity* refers to the condition when two or more of the independent variables, or linear combinations of the independent variables, in a multiple regression are highly correlated with each other. This condition distorts the standard error of estimate and the coefficient standard errors, leading to problems when conducting  $t$ -tests for statistical significance of parameters.

(Module 1.3, LOS 1.j)

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### Question #3 of 139

Question ID: 1479908

An analyst is trying to determine whether fund return performance is persistent. The analyst divides funds into three groups based on whether their return performance was in the top third (group 1), middle third (group 2), or bottom third (group 3) during the previous year. The manager then creates the following equation:  $R = a + b_1D_1 + b_2D_2 + b_3D_3 + \epsilon$ , where  $R$  is return premium on the fund (the return minus the return on the S&P 500 benchmark) and  $D_i$  is equal to 1 if the fund is in group  $i$ . Assuming no other information, this equation will suffer from:

- A) multicollinearity.
- B) serial correlation.
- C) heteroskedasticity.



#### Explanation

When we use dummy variables, we have to use one less than the states of the world. In this case, there are three states (groups) possible. We should have used only two dummy variables. Multicollinearity is a problem in this case. Specifically, a linear combination of independent variables is perfectly correlated.  $X_1 + X_2 + X_3 = 1$ .

There are too many dummy variables specified, so the equation will suffer from multicollinearity.

(Module 1.3, LOS 1.h)

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### Question #4 of 139

Question ID: 1479904

Which of the following is *least likely* to result in misspecification of a regression model?

- A) Inappropriate variable form.
- B) Transforming a variable.
- C) Omission of an important independent variable.



#### Explanation

The four types of model specification errors are: omission of an important independent variable, inappropriate variable form, inappropriate variable scaling and data improperly pooled. Transforming an independent variable is usually done to rectify inappropriate variable scaling.

(Module 1.3, LOS 1.g)

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### Question #5 of 139

Question ID: 1471891

Which of the following statements regarding the results of a regression analysis is *least* accurate? The:

- slope coefficient in a multiple regression is the change in the dependent

**A)** variable for a one-unit change in the independent variable, holding all other variables constant. 
- B)** slope coefficients in the multiple regression are referred to as partial betas. 
- C)** slope coefficient in a multiple regression is the value of the dependent variable for a given value of the independent variable. 

#### Explanation

The slope coefficient is the change in the dependent variable for a one-unit change in the independent variable.

(Module 1.1, LOS 1.b)

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### Question #6 of 139

Question ID: 1471927

One of the underlying assumptions of a multiple regression is that the variance of the residuals is constant for various levels of the independent variables. This quality is referred to as:

- A)** homoskedasticity. 
- B)** a normal distribution. 
- C)** a linear relationship. 

#### Explanation

Homoskedasticity refers to the basic assumption of a multiple regression model that the variance of the error terms is constant.

(Module 1.1, LOS 1.c)

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### Question #7 of 139

Question ID: 1471867

Consider the following estimated regression equation, with calculated  $t$ -statistics of the estimates as indicated:

$$\text{AUTO}_t = 10.0 + 1.25 \text{PI}_t + 1.0 \text{TEEN}_t - 2.0 \text{INS}_t$$

with a PI calculated  $t$ -statistic of 0.45, a TEEN calculated  $t$ -statistic of 2.2, and an INS calculated  $t$ -statistic of 0.63.

The equation was estimated over 40 companies. Using a 5% level of significance, which of the independent variables significantly different from zero?

A) TEEN only.



B) PI and INS only.



C) PI only.



#### Explanation

The critical  $t$ -values for  $40-3-1 = 36$  degrees of freedom and a 5% level of significance are  $\pm 2.028$ . Therefore, only TEEN is statistically significant.

(Module 1.1, LOS 1.b)

#### Question #8 of 139

Question ID: 1471873

Henry Hilton, CFA, is undertaking an analysis of the bicycle industry. He hypothesizes that bicycle sales (SALES) are a function of three factors: the population under 20 (POP), the level of disposable income (INCOME), and the number of dollars spent on advertising (ADV). All data are measured in millions of units. Hilton gathers data for the last 20 years and estimates the following equation (standard errors in parentheses):

|       |         |             |                |             |
|-------|---------|-------------|----------------|-------------|
| SALES | = 0.000 | + 0.004 POP | + 1.031 INCOME | + 2.002 ADV |
|       | (0.113) | (0.005)     | (0.337)        | (2.312)     |

For next year, Hilton estimates the following parameters: (1) the population under 20 will be 120 million, (2) disposable income will be \$300,000,000, and (3) advertising expenditures will be \$100,000,000. Based on these estimates and the regression equation, what are predicted sales for the industry for next year?

A) \$656,991,000.



B) \$509,980,000.



C) \$557,143,000.



### Explanation

Predicted sales for next year are:

$$\text{SALES} = \alpha + 0.004 (120) + 1.031 (300) + 2.002 (100) = 509,980,000.$$

(Module 1.1, LOS 1.b)

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### Question #9 of 139

Question ID: 1479883

One possible problem that could jeopardize the validity of the employment growth rate model is multicollinearity. Which of the following would *most likely* suggest the existence of multicollinearity?

- A) The variance of the observations has increased over time. 
- B) The Durbin–Watson statistic is significant. 
- C) The F-statistic suggests that the overall regression is significant, however the regression coefficients are not individually significant. 

### Explanation

One symptom of multicollinearity is that the regression coefficients may not be individually statistically significant even when according to the F-statistic the overall regression is significant. The problem of multicollinearity involves the existence of high correlation between two or more independent variables. Clearly, as service employment rises, construction employment must rise to facilitate the growth in these sectors. Alternatively, as manufacturing employment rises, the service sector must grow to serve the broader manufacturing sector.

- The variance of observations suggests the possible existence of heteroskedasticity.
- If the Durbin–Watson statistic may be used to test for serial correlation at a single lag.

(Module 1.2, LOS 1.f)

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### Question #10 of 139

Question ID: 1586006

Consider the following estimated regression equation:

$$\text{AUTO}_t = 10.0 + 1.25 \text{PI}_t + 1.0 \text{TEEN}_t - 2.0 \text{INS}_t$$

The equation was estimated over 40 companies. The predicted value of AUTO if PI is 4, TEEN is 0.30, and INS = 0.6 is *closest* to:

A) 14.90.



B) 17.50.



C) 14.10.



### Explanation

Predicted AUTO

$$= 10 + 1.25 (4) + 1.0 (0.30) - 2.0 (0.6)$$

$$= 10 + 5 + 0.3 - 1.2$$

$$= 14.10$$

(Module 1.2, LOS 1.f)

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Ben Sasse is a quantitative analyst at Gurnop Asset Managers. Sasse is interviewing Victor Sophie for a junior analyst position. Sasse mentions that the firm currently uses several proprietary multiple regression models and wants Sophie's opinion about regression models.

Sophie makes the following statements:

Statement 1: Multiple regression models can be used to forecast independent variables.

Statement 2: Multiple regression models can be used to test existing theories of relationships among variables.

Sasse then discusses a model that the firm uses to forecast credit spread on investment-grade corporate bonds. Sasse states that while the current model parameters are a secret, the following is an older version of the model:

$$\text{CSP} = 0.22 + 1.04 \times \text{DSC} - 0.32 \times \text{index} + 1.33 \times \text{D/E}$$

where:

CSP = credit spread (%)

DSC = EBITDA / unsecured debt

index = 1 if the issuer is part of CDX index; 0 otherwise

D/E = long-term debt / equity

Regarding Sophie's statement on multiple regression:

- A) only Statement 1 is correct.
- B) only Statement 2 is correct.
- C) both statements are correct.



#### Explanation

Multiple regression models can be used to *identify* relations between variables, forecast the *dependent* variable, and *test* existing theories. Statement 1 is inaccurate in because it mentions forecast independent (and not dependent) variables.

(Module 1.1, LOS 1.a)

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### Question #12 - 14 of 139

Question ID: 1501588

Based on the credit spread model, if an issuer gets included in the CDX index and assuming everything else the same, which of the following statements *most accurately* describes the model's forecast?

- A) The credit spread on the firm's issue would decrease by 10 bps.
- B) The credit spread on the firm's issue will increase by 32 bps.
- C) The credit spread on the firm's issue will decrease by 32 bps.



#### Explanation

The coefficient on the index dummy variable is  $-0.32$ , and if the variable takes a value of 1 (inclusion in the index), the credit spread would decrease by  $0.32\%$ , or 32 bps.

(Module 1.1, LOS 1.b)

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### Question #13 - 14 of 139

Question ID: 1501589

Which of the following is *least likely* an assumption of multiple linear regression?

- A) The dependent variable is not serially correlated.
- B) There is no linear relationship between the independent variables.
- C) The error term is normally distributed.



### Explanation

The assumption calls for the residual (or errors) to be not serially correlated. The dependent variable can have serial correlation. Other assumptions are accurate.

(Module 1.1, LOS 1.c)

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### Question #14 - 14 of 139

Question ID: 1501590

Which assumption of multiple regression is *most likely* evaluated using a QQ plot?

- A) Serial correlation of residuals.
- B) Conditional heteroskedasticity.
- C) Error term is normally distributed.



### Explanation

A normal QQ plot of the residuals can visually indicate violation of the assumption that the residuals are normally distributed.

(Module 1.1, LOS 1.c)

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### Question #15 of 139

Question ID: 1471872

Henry Hilton, CFA, is undertaking an analysis of the bicycle industry. He hypothesizes that bicycle sales (SALES) are a function of three factors: the population under 20 (POP), the level of disposable income (INCOME), and the number of dollars spent on advertising (ADV). All data are measured in millions of units. Hilton gathers data for the last 20 years and estimates the following equation (standard errors in parentheses):

|       |            |             |                |             |
|-------|------------|-------------|----------------|-------------|
| SALES | = $\alpha$ | + 0.004 POP | + 1.031 INCOME | + 2.002 ADV |
|       |            | (0.005)     | (0.337)        | (2.312)     |

The critical t-statistic for a 95% confidence level is 2.120. Which of the independent variables is statistically different from zero at the 95% confidence level?

- A) INCOME and ADV.
- B) ADV only.





C) INCOME only.



### Explanation

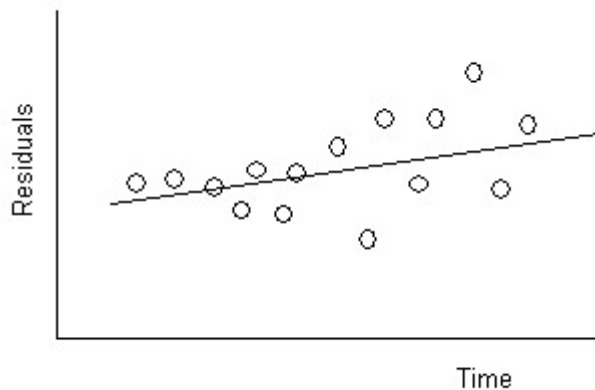
The calculated test statistic is coefficient/standard error. Hence, the t-stats are 0.8 for POP, 3.059 for INCOME, and 0.866 for ADV. Since the t-stat for INCOME is the only one greater than the critical t-value of 2.120, only INCOME is significantly different from zero.

(Module 1.1, LOS 1.b)

### Question #16 of 139

Question ID: 1472011

Consider the following graph of residuals and the regression line from a time-series regression:



These residuals exhibit the regression problem of:

A) autocorrelation.



B) homoskedasticity.



C) heteroskedasticity.



### Explanation

The residuals appear to be from two different distributions over time. In the earlier periods, the model fits rather well compared to the later periods.

(Module 1.3, LOS 1.h)

Using a recent analysis of salaries (in \$1,000) of financial analysts, Timbadia runs a regression of salaries on education, experience, and gender. (Gender equals one for men and zero for women.) The regression results from a sample of 230 financial analysts are presented below, with *t*-statistics in parenthesis.

$$\text{Salary} = 34.98 + 1.2 \text{ Education} + 0.5 \text{ Experience} + 6.3 \text{ Gender}$$

$$(29.11) \quad (8.93) \quad (2.98) \quad (1.58)$$

Timbadia also runs a multiple regression to gain a better understanding of the relationship between lumber sales, housing starts, and commercial construction. The regression uses a large data set of lumber sales as the dependent variable with housing starts and commercial construction as the independent variables. The results of the regression are:

|                         | Coefficient | Standard Error | t-statistics |
|-------------------------|-------------|----------------|--------------|
| Intercept               | 5.337       | 1.71           | 3.14         |
| Housing starts          | 0.76        | 0.09           | 8.44         |
| Commercial construction | 1.25        | 0.33           | 3.78         |

Finally, Timbadia runs a regression between the returns on a stock and its industry index with the following results:

|                | Coefficient | Standard Error |
|----------------|-------------|----------------|
| Intercept      | 2.1         | 2.01           |
| Industry index | 1.9         | 0.31           |

- Standard error of estimate = 15.1
- Correlation coefficient = 0.849

### Question #17 - 19 of 139

Question ID: 1479889

What is the expected salary (in \$1,000) of a woman with 16 years of education and 10 years of experience?

A) 65.48.



B) 59.18.



C) 54.98.



#### Explanation

$$34.98 + 1.2(16) + 0.5(10) = 59.18$$

(Module 1.2, LOS 1.f)

### Question #18 - 19 of 139

Question ID: 1479890

If the return on the industry index is 4%, the stock's expected return would be:

A) 9.7%.



B) 7.6%.



C) 11.2%.



#### Explanation

$$Y = b_0 + bX_1$$

$$Y = 2.1 + 1.9(4) = 9.7\%$$

(Module 1.2, LOS 1.f)

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### Question #19 - 19 of 139

Question ID: 1479891

The percentage of the variation in the stock return explained by the variation in the industry index return is *closest* to:

A) 72.1%.



B) 63.2%.



C) 84.9%.



#### Explanation

The coefficient of determination,  $R^2$ , is the square the correlation coefficient.  $0.849^2 = 0.721$ .

(Module 1.2, LOS 1.d)

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### Question #20 of 139

Question ID: 1471980

An analyst runs a regression of monthly value-stock returns on five independent variables over 48 months. The total sum of squares is 430, and the sum of squared errors is 170. Test the null hypothesis at the 2.5% and 5% significance level that all five of the independent variables are equal to zero.

A) Rejected at 5% significance only.



B) Not rejected at 2.5% or 5.0% significance.



C) Rejected at 2.5% significance and 5% significance.



### Explanation

The F-statistic is equal to the ratio of the mean squared regression (MSR) to the mean squared error (MSE).

$$RSS = SST - SSE = 430 - 170 = 260$$

$$MSR = 260 / 5 = 52$$

$$MSE = 170 / (48 - 5 - 1) = 4.05$$

$$F = 52 / 4.05 = 12.84$$

The critical F-value for 5 and 42 degrees of freedom at a 5% significance level is approximately 2.44. The critical F-value for 5 and 42 degrees of freedom at a 2.5% significance level is approximately 2.89. Therefore, we can reject the null hypothesis at either level of significance and conclude that at least one of the five independent variables explains a significant portion of the variation of the dependent variable.

(Module 1.2, LOS 1.e)

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### Question #21 of 139

Question ID: 1472075

Jill Wentraub is an analyst with the retail industry. She is modeling a company's sales over time and has noticed a quarterly seasonal pattern. If she includes dummy variables to represent the seasonality component of the sales she must use:

A) three dummy variables.



B) four dummy variables.



C) one dummy variables.



### Explanation

Three. Always use one less dummy variable than the number of possibilities. For a seasonality that varies by quarters in the year, three dummy variables are needed.

(Module 1.4, LOS 1.I)

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### Question #22 of 139

Question ID: 1471947

An analyst regresses the return of a S&P 500 index fund against the S&P 500, and also regresses the return of an active manager against the S&P 500. The analyst uses the last five years of data in both regressions. Without making any other assumptions, which of the following is *most* accurate? The index fund:

- A) regression should have higher sum of squares regression as a ratio to the total sum of squares. 
- B) should have a higher coefficient on the independent variable. 
- C) should have a lower coefficient of determination. 

#### Explanation

The index fund regression should provide a higher  $R^2$  than the active manager regression.  $R^2$  is the sum of squares regression divided by the total sum of squares.

(Module 1.2, LOS 1.d)

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### Question #23 of 139

Question ID: 1479923

Assume that in a particular multiple regression model, it is determined that the error terms are uncorrelated with each other. Which of the following statements is *most* accurate?

- A) This model is in accordance with the basic assumptions of multiple regression analysis because the errors are not serially correlated. 
- B) Unconditional heteroskedasticity present in this model should not pose a problem, but can be corrected by using robust standard errors. 
- C) Serial correlation may be present in this multiple regression model, and can be confirmed only through a Durbin-Watson test. 

#### Explanation

One of the basic assumptions of multiple regression analysis is that the error terms are not correlated with each other. In other words, the error terms are not serially correlated. Multicollinearity and heteroskedasticity are problems in multiple regression that are not related to the correlation of the error terms.

(Module 1.3, LOS 1.i)

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Vijay Shapule, CFA, is investigating the application of the Fama-French three-factor model (Model 1) for the Indian stock market for the period 2001–2011 (120 months). Using the dependent variable as annualized return (%), the results of the analysis are shown in **Indian Equities—Fama-French Model**.

### Indian Equities—Fama-French Model

| Factor    | Coefficient | P-Value |
|-----------|-------------|---------|
| Intercept | 1.22        | <0.001  |
| SMB       | 0.23        | <0.001  |
| HML       | 0.34        | 0.003   |
| Rm-Rf     | 0.88        | <0.001  |
| R-squared |             | 0.36    |
| SSE       |             | 38.00   |
| AIC       |             | -129.99 |
| BIC       |             | -118.84 |

Shapule then modifies the model to include a liquidity factor. Results for this four-factor model (Model 2) are shown in **Revised Fama-French Model With Liquidity Factor**

### Revised Fama-French Model With Liquidity Factor

| Factor    | Coefficient | P-Value |
|-----------|-------------|---------|
| Intercept | 1.56        | <0.001  |
| SMB       | 0.22        | <0.001  |
| HML       | 0.35        | 0.012   |
| Rm-Rf     | 0.87        | <0.001  |
| LIQ       | -0.12       | 0.02    |
| R-squared |             | 0.39    |
| SSE       |             | 34.00   |
| AIC       |             | -141.34 |
| BIC       |             | -127.40 |

### Question #24 - 27 of 139

Question ID: 1501592

The adjusted  $R^2$  of Model 2 is *closest* to:

A) 0.39.



B) 0.37.



C) 0.36.



### Explanation

Given  $n = 120$  months,  $k = 4$  (for Model 2), and  $R^2 = 0.39$ :

$$R_a^2 = 1 - \left[ \left( \frac{120-1}{120-4-1} \right) \times (1 - 0.39) \right] = 0.37$$

(Module 1.2, LOS 1.d)

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### Question #25 - 27 of 139

Question ID: 1501593

The model better suited for prediction is:

A) Model 1 because it has a lower Bayesian information criterion.



B) Model 2 because it has a higher Akaike information criterion.



C) Model 2 because it has a lower Akaike information criterion.



### Explanation

The Akaike information criterion (AIC) is used if the goal is to have a better forecast, while the Bayesian information criterion (BIC) is used if the goal is a better goodness of fit. Lower values of both criteria indicate a better model. Both criteria are lower for Model 2.

(Module 1.2, LOS 1.d)

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### Question #26 - 27 of 139

Question ID: 1501594

The  $F$ -statistic for testing  $H_0$ : coefficient of LIQ = 0 versus  $H_a$ : coefficient of LIQ  $\neq$  0 is *closest* to:

A) 13.33.



B) 5.45.



C) 2.11.



### Explanation

$$F = \frac{(SSE_R - SSE_U)/q}{(SSE_U)/(n-k-1)}$$

where  $n = 120$ ,  $k = 4$ , and  $q = 1$ :

$$\frac{(38-34)/1}{(34)/(120-4-1)} = 13.33$$

(Module 1.2, LOS 1.d)

### Question #27 - 27 of 139

Question ID: 1501595

What is the predicted return for a stock using Model 1 when  $SMB = 3.30$ ,  $HML = 1.25$  and  $R_m - R_f = 5$ ?

**A)** 6.80%.



**B)** 7.88%.



**C)** 9.58%.



#### Explanation

Model 1:

$$\begin{aligned} \text{Return} &= 1.22 + 0.23 \times SMB + 0.34 \times HML + 0.88 R_m - R_f \\ &= 1.22 + 0.23 \times 3.30 + 0.34 \times 1.25 + 0.88 \times 5 \\ &= 6.80\%. \end{aligned}$$

(Module 1.2, LOS 1.d)

### Question #28 of 139

Question ID: 1479901

A multiple regression model has included independent variables that are not linearly related to the dependent variable. The model is *most likely* misspecified due to:

**A)** incorrect data pooling.



**B)** incorrect variable form.



**C)** incorrect variable scaling.





### Explanation

Incorrect variable form misspecification occurs if the relationship between dependent and independent variables is nonlinear.

(Module 1.3, LOS 1.g)

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### Question #29 of 139

Question ID: 1472077

Consider the following model of earnings (EPS) regressed against dummy variables for the quarters:

$$EPS_t = \alpha + \beta_1 Q_{1t} + \beta_2 Q_{2t} + \beta_3 Q_{3t}$$

where:

$EPS_t$  is a quarterly observation of earnings per share

$Q_{1t}$  takes on a value of 1 if period  $t$  is the second quarter, 0 otherwise

$Q_{2t}$  takes on a value of 1 if period  $t$  is the third quarter, 0 otherwise

$Q_{3t}$  takes on a value of 1 if period  $t$  is the fourth quarter, 0 otherwise

Which of the following statements regarding this model is *most* accurate? The:

- A) significance of the coefficients cannot be interpreted in the case of dummy variables. 
- B) coefficient on each dummy tells us about the difference in earnings per share between the respective quarter and the one left out (first quarter in this case). 
- C) EPS for the first quarter is represented by the residual. 

### Explanation

The coefficients on the dummy variables indicate the difference in EPS for a given quarter, relative to the first quarter.

(Module 1.4, LOS 1.I)

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### Question #30 of 139

Question ID: 1479913

One of the main assumptions of a multiple regression model is that the variance of the residuals is constant across all observations in the sample. A violation of the assumption is *most likely* to be described as:

A) positive serial correlation.



B) heteroskedasticity.



C) unstable remnant deviation.



#### Explanation

Heteroskedasticity is present when the variance of the residuals is not the same across all observations in the sample, and there are sub-samples that are more spread out than the rest of the sample.

(Module 1.3, LOS 1.h)

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### Question #31 of 139

Question ID: 1472012

Which of the following statements regarding heteroskedasticity is *least* accurate?

A) Heteroskedasticity may occur in cross-sectional or time-series analyses.



B) The assumption of linear regression is that the residuals are heteroskedastic.



C) Heteroskedasticity results in an estimated variance that is too small and, therefore, affects statistical inference.



#### Explanation

The assumption of regression is that the residuals are homoskedastic (i.e., the residuals are drawn from the same distribution).

(Module 1.3, LOS 1.h)

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Werner Baltz, CFA, has regressed 30 years of data to forecast future sales for National Motor Company based on the percent change in gross domestic product (GDP) and the change in retail price of a U.S. gallon of fuel. The results are presented below.

| Predictor        | Coefficient | Standard Error of the Coefficient |
|------------------|-------------|-----------------------------------|
| Intercept        | 78          | 13.710                            |
| $\Delta$ GDP     | 30.22       | 12.120                            |
| $\Delta$ \$ Fuel | -412.39     | 183.981                           |

| Analysis of Variance Table (ANOVA) |                    |                |
|------------------------------------|--------------------|----------------|
| Source                             | Degrees of Freedom | Sum of Squares |
| Regression                         |                    | 291.30         |
| Error                              | 27                 | 132.12         |
| Total                              | 29                 | 423.42         |

### Question #32 - 34 of 139

Question ID: 1479910

If GDP rises 2.2% and the price of fuels falls \$0.15, Baltz's model will predict Company sales to be (in \$ millions) *closest* to:

- A) \$82.00. ✗
- B) \$128.00. ✗
- C) \$206.00. ✓

#### Explanation

Sales will be closest to  $\$78 + (\$30.22 \times 2.2) + [(-412.39) \times (-\$0.15)] = \$206.34$  million.

(Module 1.2, LOS 1.f)

### Question #33 - 34 of 139

Question ID: 1479911

Baltz proceeds to test the hypothesis that none of the independent variables has significant explanatory power. He concludes that, at a 5% level of significance:

- A) all of the independent variables have explanatory power, because the calculated  $F$ -statistic exceeds its critical value. ✗
- B) none of the independent variables has explanatory power, because the calculated  $F$ -statistic does not exceed its critical value. ✗
- C) at least one of the independent variables has explanatory power, because the calculated  $F$ -statistic exceeds its critical value. ✓

#### Explanation

$MSE = SSE / [n - (k + 1)] = 132.12 \div 27 = 4.89$ . From the ANOVA table, the calculated  $F$ -statistic is (mean square regression / mean square error) =  $145.65 / 4.89 = 29.7853$ . From the  $F$ -distribution table (2 df numerator, 27 df denominator) the  $F$ -critical value may be interpolated to be 3.36. Because 29.7853 is greater than 3.36, Baltz rejects the null hypothesis and concludes that at least one of the independent variables has explanatory power.

(Module 1.2, LOS 1.e)

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### Question #34 - 34 of 139

Question ID: 1479912

Presence of conditional heteroskedasticity is *least likely* to affect the:

- A) computed  $F$ -statistic. 
- B) coefficient estimates. 
- C) computed  $t$ -statistic. 

#### Explanation

Conditional heteroskedasticity results in consistent coefficient estimates, but it biases standard errors, affecting the computed  $t$ -statistic and  $F$ -statistic.

(Module 1.3, LOS 1.h)

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### Question #35 of 139

Question ID: 1471928

Which of the following statements *least* accurately describes one of the fundamental multiple regression assumptions?

- A) The error term is normally distributed. 
- B) The independent variables are not random. 
- C) The variance of the error terms is not constant (i.e., the errors are heteroskedastic). 

#### Explanation

The variance of the error term IS assumed to be constant, resulting in errors that are homoskedastic.

(Module 1.1, LOS 1.c)

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**Question #36 of 139**

Question ID: 1479878

Consider the following analysis of variance table:

| Source     | Sum of Squares | Df | Mean Square |
|------------|----------------|----|-------------|
| Regression | 20             | 1  | 20          |
| Error      | 80             | 20 | 4           |
| Total      | 100            | 21 |             |

The  $F$ -statistic for a test of joint significance of all the slope coefficients is *closest* to:

A) 0.2.



B) 5.



C) 0.05.

**Explanation**

The  $F$ -statistic is equal to the ratio of the mean squared regression to the mean squared error.

$$F = MSR / MSE = 20 / 4 = 5.$$

(Module 1.2, LOS 1.e)

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William Brent, CFA, is the chief financial officer for Mega Flowers, one of the largest producers of flowers and bedding plants in the Western United States. Mega Flowers grows its plants in three large nursery facilities located in California. Its products are sold in its company-owned retail nurseries as well as in large, home and garden "super centers". For its retail stores, Mega Flowers has designed and implemented marketing plans each season that are aimed at its consumers in order to generate additional sales for certain high-margin products. To fully implement the marketing plan, additional contract salespeople are seasonally employed.

For the past several years, these marketing plans seemed to be successful, providing a significant boost in sales to those specific products highlighted by the marketing efforts. However, for the past year, revenues have been flat, even though marketing expenditures increased slightly. Brent is concerned that the expensive seasonal marketing campaigns are simply no longer generating the desired returns, and should either be significantly modified or eliminated altogether. He proposes that the company hire additional, permanent salespeople to focus on selling Mega Flowers' high-margin products all year long. The chief

operating officer, David Johnson, disagrees with Brent. He believes that although last year's results were disappointing, the marketing campaign has demonstrated impressive results for the past five years, and should be continued. His belief is that the prior years' performance can be used as a gauge for future results, and that a simple increase in the sales force will not bring about the desired results.

Brent gathers information regarding quarterly sales revenue and marketing expenditures for the past five years. Based upon historical data, Brent derives the following regression equation for Mega Flowers (stated in millions of dollars):

$$\text{Expected Sales} = 12.6 + 1.6 (\text{Marketing Expenditures}) + 1.2 (\# \text{ of Salespeople})$$

Brent shows the equation to Johnson and tells him, "This equation shows that a \$1 million increase in marketing expenditures will increase the independent variable by \$1 .6 million, all other factors being equal." Johnson replies , "It also appears that sales will equal \$12.6 million if all independent variables are equal to zero."

Brent makes the following statements about model evaluation:

- Statement 1: The BIC metric usually imposes a higher penalty for overfitting than AIC.
- Statement 2: AIC is used if the goal is to have a better forecast, while BIC is used if the goal is a better goodness of fit.

### Question #37 - 40 of 139

Question ID: 1471900

In regard to their conversation about the regression equation:

- A)** Brent's statement is correct; Johnson's statement is incorrect. 
- B)** Brent's statement is correct; Johnson's statement is correct. 
- C)** Brent's statement is incorrect; Johnson's statement is correct. 

#### Explanation

Expected sales is the dependent variable in the equation, while expenditures for marketing and salespeople are the independent variables. Therefore, a \$1 million increase in marketing expenditures will increase the *dependent* variable (expected sales) by \$1.6 million. Brent's statement is incorrect.

Johnson's statement is correct. 12.6 is the intercept in the equation, which means that if all independent variables are equal to zero, expected sales will be \$12.6 million.

(Module 1.1, LOS 1.b)

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### Question #38 - 40 of 139

Question ID: 1586001

Regarding Brent's Statements 1 and 2:

- A) Both statements are correct.
- B) Only Statement 2 is correct.
- C) Only Statement 1 is correct.



#### Explanation

Statement 1 is correct. Comparing the formulae for computation of AIC and BIC, because  $\ln(n)$  is greater than 2 for even small sample sizes, the BIC metric imposes a higher penalty for overfitting. Statement 2 is correct. Both AIC and BIC evaluate the quality of model fit *among competing models for the same dependent variable*. Lower values indicate a better model under either criteria. AIC is used if the goal is to have a better forecast, while BIC is used if the goal is a better goodness of fit.

(Module 1.2, LOS 1.d)

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### Question #39 - 40 of 139

Question ID: 1471904

Assuming that next year's marketing expenditures are \$3,500,000 and there are five salespeople, predicted sales for Mega Flowers should will be:

- A) \$11,600,000.
- B) \$24,000,000.
- C) \$24,200,000.



#### Explanation

Using the information provided, expected sales equals  $12.6 + (1.6 \times 3.5) + (1.2 \times 5) = \$24.2$  million. Remember to check the details - i.e. this equation is denominated in millions of dollars.

(Module 1.1, LOS 1.b)

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### Question #40 - 40 of 139

Question ID: 1471905

Brent would like to further investigate whether at least one of the independent variables can explain a significant portion of the variation of the dependent variable. Which of the following methods would be best for Brent to use?

**A)** The  $F$ -statistic.



**B)** The multiple coefficient of determination.



**C)** An ANOVA table.



### Explanation

To determine whether at least one of the coefficients is statistically significant, the calculated  $F$ -statistic is compared with the critical  $F$ -value at the appropriate level of significance.

(Module 1.1, LOS 1.b)

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In preparing an analysis of HB Inc., Jack Stumper is asked to look at the company's sales in relation to broad based economic indicators. Stumper's analysis indicates that HB's monthly sales are related to changes in housing starts (H) and changes in the mortgage interest rate (M). The analysis covers the past ten years for these variables. The regression equation is:

$$S = 1.76 + 0.23H - 0.08M$$

|                           |       |
|---------------------------|-------|
| Number of observations:   | 123   |
| Unadjusted $R^2$ :        | 0.77  |
| F statistic:              | 9.80  |
| Durbin Watson statistic   | 0.50  |
| p-value of Housing Starts | 0.017 |
| p=value of Mortgage Rates | 0.033 |

### Variable Descriptions

S = HB Sales (in thousands)

H = housing starts (in thousands)

M = mortgage interest rate (in percent)

### November 20x6 Actual Data

HB's monthly sales: \$55,000

Housing starts: 150,000

Mortgage interest rate (%): 7.5



**Question #41 - 46 of 139**

Question ID: 1471914

Using the regression model developed, the closest prediction of sales for December 20x6 is:

- A) \$55,000. 
- B) \$36,000. 
- C) \$44,000. 

**Explanation**

$$1.76 + 0.23 * (150) - 0.08 * (7.5) = 35.66.$$

(Module 1.1, LOS 1.b)

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**Question #42 - 46 of 139**

Question ID: 1471915

Will Stumper conclude that the housing starts coefficient is statistically different from zero and how will he interpret it at the 5% significance level:

- A) different from zero; sales will rise by \$100 for every 23 house starts. 
- B) not different from zero; sales will rise by \$0 for every 100 house starts. 
- C) different from zero; sales will rise by \$23 for every 100 house starts. 

**Explanation**

A p-value (0.017) below significance (0.05) indicates a variable which is statistically different from zero. The coefficient of 0.23 indicates that sales will rise by \$23 for every 100 house starts.

(Module 1.1, LOS 1.b)

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**Question #43 - 46 of 139**

Question ID: 1585998

Is the regression coefficient of changes in mortgage interest rates different from zero at the 5 percent level of significance?

- A) yes, because p-value < 0.05. 
- B) no, because coefficient is negative. 
- C) yes, because  $-0.08 < 0.05$ . 

**Explanation**

A coefficient is statistically significantly different from zero if its p-value is less than the level of significance.

(Module 1.1, LOS 1.b)

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### Question #44 - 46 of 139

Question ID: 1471917

In this multiple regression, the F-statistic indicates the:

- A) deviation of the estimated values from the actual values of the dependent variable. 
- B) the joint significance of the independent variables. 
- C) degree of correlation between the independent variables. 

#### Explanation

The F-statistic indicates the joint significance of the independent variables. The deviation of the estimated values from the actual values of the dependent variable is the standard error of estimate. The degree of correlation between the independent variables is the coefficient of correlation.

(Module 1.1, LOS 1.b)

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### Question #45 - 46 of 139

Question ID: 1471918

The regression statistics above indicate that for the period under study, the independent variables (housing starts, mortgage interest rate) together explained approximately what percentage of the variation in the dependent variable (sales)?

- A) 77.00. 
- B) 9.80. 
- C) 67.00. 

#### Explanation

The question is asking for the coefficient of determination.

(Module 1.1, LOS 1.b)

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In this multiple regression, if Stumper discovers that the residuals exhibit positive serial correlation, the *most likely* effect is:

- A) standard errors are too low but coefficient estimate is consistent. 
- B) standard errors are too high but coefficient estimate is consistent. 
- C) standard errors are not affected but coefficient estimate is inconsistent. 

### Explanation

Positive serial correlation in residuals does not affect the consistency of coefficients (i.e., the coefficients are still consistent) but the estimated standard errors are too low leading to artificially high t-statistics.

(Module 1.1, LOS 1.b)

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George Smith, an analyst with Great Lakes Investments, has created a comprehensive report on the pharmaceutical industry at the request of his boss. The Great Lakes portfolio currently has a significant exposure to the pharmaceuticals industry through its large equity position in the top two pharmaceutical manufacturers. His boss requested that Smith determine a way to accurately forecast pharmaceutical sales in order for Great Lakes to identify further investment opportunities in the industry as well as to minimize their exposure to downturns in the market. Smith realized that there are many factors that could possibly have an impact on sales, and he must identify a method that can quantify their effect. Smith used a multiple regression analysis with five independent variables to predict industry sales. His goal is to not only identify relationships that are statistically significant, but economically significant as well. The assumptions of his model are fairly standard: a linear relationship exists between the dependent and independent variables, the independent variables are not random, and the expected value of the error term is zero.

Smith is confident with the results presented in his report. He has already done some hypothesis testing for statistical significance, including calculating a t-statistic and conducting a two-tailed test where the null hypothesis is that the regression coefficient is equal to zero versus the alternative that it is not. He feels that he has done a thorough job on the report and is ready to answer any questions posed by his boss.

However, Smith's boss, John Sutter, is concerned that in his analysis, Smith has ignored several potential problems with the regression model that may affect his conclusions. He knows that when any of the basic assumptions of a regression model are violated, any results drawn for the model are questionable. He asks Smith to go back and carefully examine the effects of heteroskedasticity, multicollinearity, and serial correlation on his

model. In specific, he wants Smith to make suggestions regarding how to detect these errors and to correct problems that he encounters.

### Question #47 - 50 of 139

Question ID: 1479940

Sutter has detected the presence of conditional heteroskedasticity in Smith's report. This is evidence that:

- A) the error terms are correlated with each other. 
- B) the variance of the error term is correlated with the values of the independent variables. 
- C) two or more of the independent variables are highly correlated with each other. 

#### Explanation

Conditional heteroskedasticity exists when the variance of the error term is correlated with the values of the independent variables.

Multicollinearity, on the other hand, occurs when two or more of the independent variables are highly correlated with each other. Serial correlation exists when the error terms are correlated with each other.

(Module 1.3, LOS 1.j)

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### Question #48 - 50 of 139

Question ID: 1479941

Suppose there is evidence that the variance of the error term is correlated with the values of the independent variables. The *most likely* effect on the statistical inferences Smith can make from the regressions results using financial data is to commit a:

- A) Type I error by incorrectly failing to reject the null hypothesis that the regression parameters are equal to zero. 
- B) Type II error by incorrectly failing to reject the null hypothesis that the regression parameters are equal to zero. 
- C) Type I error by incorrectly rejecting the null hypotheses that the regression parameters are equal to zero. 

#### Explanation

One problem with conditional heteroskedasticity while working with financial data, is that the standard errors of the parameter estimates will be too small and the t-statistics too large. This will lead Smith to incorrectly reject the null hypothesis that the parameters are equal to zero. In other words, Smith will incorrectly conclude that the parameters are statistically significant when in fact they are not. This is an example of a Type I error: incorrectly rejecting the null hypothesis when it should not be rejected.

(Module 1.3, LOS 1.h)

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### Question #49 - 50 of 139

Question ID: 1479942

Which of the following is *most likely* to indicate that two or more of the independent variables, or linear combinations of independent variables, may be highly correlated with each other? Unless otherwise noted, significant and insignificant mean significantly different from zero and not significantly different from zero, respectively.

- A) The  $R^2$  is low, the F-statistic is insignificant and the Durbin-Watson statistic is significant. 
- B) The  $R^2$  is high, the F-statistic is significant and the t-statistics on the individual slope coefficients are insignificant. 
- C) The  $R^2$  is high, the F-statistic is significant and the t-statistics on the individual slope coefficients are significant. 

#### Explanation

Multicollinearity occurs when two or more of the independent variables, or linear combinations of independent variables, may be highly correlated with each other. In a classic effect of multicollinearity, the  $R^2$  is high and the F-statistic is significant, but the t-statistics on the individual slope coefficients are insignificant.

(Module 1.3, LOS 1.j)

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### Question #50 - 50 of 139

Question ID: 1479943

Using the Durbin-Watson test statistic, Smith rejects the null hypothesis suggested by the test. This is evidence that:

- A) two or more of the independent variables are highly correlated with each other. 

**B)** the error term is normally distributed.



**C)** the error terms are correlated with each other.



### Explanation

Serial correlation (also called autocorrelation) exists when the error terms are correlated with each other.

Multicollinearity, on the other hand, occurs when two or more of the independent variables are highly correlated with each other. One assumption of multiple regression is that the error term is normally distributed.

(Module 1.3, LOS 1.i)

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### Question #51 of 139

Question ID: 1479949

Suppose the analyst wants to add a dummy variable for whether a person has a business college degree and an engineering degree. What is the *CORRECT* representation if a person has both degrees?

| <u>Business</u>     | <u>Engineering</u>  |
|---------------------|---------------------|
| <u>Degree Dummy</u> | <u>Degree Dummy</u> |
| <u>Variable</u>     | <u>Variable</u>     |

**A)** 0                      0



**B)** 1                      1



**C)** 0                      1



### Explanation

Assigning a zero to both categories is appropriate for someone with neither degree. Assigning one to the business category and zero to the engineering category is appropriate for someone with only a business degree. Assigning zero to the business category and one to the engineering category is appropriate for someone with only an engineering degree. Assigning a one to both categories is correct because it reflects the possession of both degrees.

(Module 1.4, LOS 1.i)

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### Question #52 of 139

Question ID: 1479921

Alex Wade, CFA, is analyzing the result of a regression analysis comparing the performance of gold stocks versus a broad equity market index. Wade believes that first lag serial correlation may be present and, in order to prove his theory, should use which of the following methods to detect its presence?

**A)** The Breusch-Pagan test.



**B)** The Hansen method.



**C)** The Durbin-Watson statistic.



### Explanation

The Durbin-Watson statistic is the most commonly used method for the detection of serial correlation at the first lag, although residual plots can also be utilized. For testing of serial correlation beyond the first lag, we can instead use the Breusch-Godfrey test (but is not one of the answer choices).

(Module 1.3, LOS 1.i)

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Phillip Lee works for Song Bank as a quantitative analyst. He is currently working on a model to explain the returns (in %) of 20 hedge funds for the past year. He includes three independent variables:

- Market return = return on a broad-based stock index (in %)
- Closed = dummy variable (= 1 if the fund is closed to new investors; 0 otherwise)
- Prior period alpha = fund return for the prior 12 months – return on market (in %)

Estimated model: hedge fund return =  $3.2 + 0.22 \text{ market return} + 1.65 \text{ closed} - 0.11 \text{ prior period alpha}$

Lee is concerned about the impact of outliers on the estimated regression model and collects the following information:

| Observation | 1     | 2     | 3     | 4     | 5     | 6     | 7     | 8     | 9     | 10    |
|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| Cook's D    | 0.332 | 0.219 | 0.115 | 0.212 | 0.376 | 0.232 | 0.001 | 0.001 | 0.233 | 0.389 |
| Observation | 11    | 12    | 13    | 14    | 15    | 16    | 17    | 18    | 19    | 20    |
| Cook's D    | 0.089 | 0.112 | 0.001 | 0.001 | 0.219 | 0.001 | 0.112 | 0.044 | 0.517 | 0.212 |

Additionally, Lee wants to estimate the probability of a hedge fund closing to new investors, and he uses two variables:

- Fund size = log of assets under management
- Prior period alpha (defined earlier)

Results are shown as follows:

| Variable           | Coefficient |
|--------------------|-------------|
| Intercept          | -3.76       |
| Fund size          | -2.98       |
| Prior period alpha | -2.99       |

### Question #53 - 55 of 139

Question ID: 1501602

What is the correct interpretation of the coefficient of *closed* in the first regression?

- A) If a model is closed to new investors, the expected excess fund return is 1.65%. 
- B) A closed fund is likely to generate a return of 1.65%. 
- C) A closed fund is estimated to have an extra return of 1.65% relative to funds that are not closed. 

#### Explanation

The interpretation of the coefficient is the extra return relative to the alternative outcome.

(Module 1.4, LOS 1.l)

### Question #54 - 55 of 139

Question ID: 1501603

To check for only the outliers in the sample, Lee should *most appropriately* use:

- A) Studentized residuals. 
- B) leverage. 
- C) Cook's D. 

#### Explanation

Studentized residuals are used to identify outliers (in the dependent variable). Leverage is used to identify high-leverage observations (in the independent variable), while Cook's D is a composite measure (combines both independent and dependent variables) to identify influential observations.

(Module 1.4, LOS 1.k)



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**Question #55 - 55 of 139**

Question ID: 1501604

Which observations, when excluded, cause a significant change to model coefficients?

A) Observations 1, 10, and 11.



B) Observations 10 and 19.



C) Observation 19.

**Explanation**

Influential observations are those that, when excluded, cause a significant change to the model coefficients.

$$\text{Observations where Cook's } D > \sqrt{\frac{k}{n}} = \sqrt{\frac{3}{20}} = 0.3873.$$

Observations 10 ( $D = 0.389$ ) and 19 ( $D = 0.517$ ) satisfy this criteria.

(Module 1.4, LOS 1.k)

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Toni Williams, CFA, has determined that commercial electric generator sales in the Midwest U.S. for Self-Start Company is a function of several factors in each area: the cost of heating oil, the temperature, snowfall, and housing starts. Using data for the most currently available year, she runs a cross-sectional regression where she regresses the deviation of sales from the historical average in each area on the deviation of each explanatory variable from the historical average of that variable for that location. She feels this is the most appropriate method since each geographic area will have different average values for the inputs, and the model can explain how current conditions explain how generator sales are higher or lower from the historical average in each area. In summary, she regresses current sales for each area minus its respective historical average on the following variables for each area.

- The difference between the retail price of heating oil and its historical average.
- The mean number of degrees the temperature is below normal in Chicago.
- The amount of snowfall above the average.
- The percentage of housing starts above the average.

Williams used a sample of 26 observations obtained from 26 metropolitan areas in the Midwest U.S. The results are in the tables below. The dependent variable is in sales of generators in millions of dollars.

| Coefficient Estimates Table |                       |                                   |
|-----------------------------|-----------------------|-----------------------------------|
| Variable                    | Estimated Coefficient | Standard Error of the Coefficient |
| Intercept                   | 5.00                  | 1.850                             |
| \$ Heating Oil              | 2.00                  | 0.827                             |
| Low Temperature             | 3.00                  | 1.200                             |
| Snowfall                    | 10.00                 | 4.833                             |
| Housing Starts              | 5.00                  | 2.333                             |

| Analysis of Variance Table (ANOVA) |                    |                |             |
|------------------------------------|--------------------|----------------|-------------|
| Source                             | Degrees of Freedom | Sum of Squares | Mean Square |
| Regression                         | 4                  | 335.20         | 83.80       |
| Error                              | 21                 | 606.40         | 28.88       |
| Total                              | 25                 | 941.60         |             |

### Table of the F-Distribution

Critical values for right-hand tail area equal to 0.05

Numerator: df1 and Denominator: df2

|     | df1    |        |        |        |        |
|-----|--------|--------|--------|--------|--------|
| df2 | 1      | 2      | 4      | 10     | 20     |
| 1   | 161.45 | 199.50 | 224.58 | 241.88 | 248.01 |
| 2   | 18.513 | 19.000 | 19.247 | 19.396 | 19.446 |
| 4   | 7.7086 | 6.9443 | 6.3882 | 5.9644 | 5.8025 |
| 10  | 4.9646 | 4.1028 | 3.4780 | 2.9782 | 2.7740 |
| 20  | 4.3512 | 3.4928 | 2.8661 | 2.3479 | 2.1242 |

One of her goals is to forecast the sales of the Chicago metropolitan area next year. For that area and for the upcoming year, Williams obtains the following projections: heating oil prices will be \$0.10 above average, the temperature in Chicago will be 5 degrees below normal, snowfall will be 3 inches above average, and housing starts will be 3% below average.

In addition to making forecasts and testing the significance of the estimated coefficients, she plans to perform diagnostic tests to verify the validity of the model's results.

### Question #56 - 60 of 139

Question ID: 1471970

According to the model and the data for the Chicago metropolitan area, the forecast of generator sales is:

- A) \$55 million above average.
- B) \$35.2 million above the average.
- C) \$65 million above the average.



#### Explanation

The model uses a multiple regression equation to predict sales by multiplying the estimated coefficient by the observed value to get:

$$[5 + (2 \times 0.10) + (3 \times 5) + (10 \times 3) + (5 \times (-3))] \times \$1,000,000 = \$35.2 \text{ million.}$$

(Module 1.2, LOS 1.e)

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### Question #57 - 60 of 139

Question ID: 1479876

Williams proceeds to test the hypothesis that none of the independent variables has significant explanatory power. Using the joint  $F$ -test for the significance of all slope coefficients, at a 5% level of significance:

- A) none of the independent variables has explanatory power.
- B) all of the independent variables have explanatory power.
- C) at least one of the independent variables has explanatory power.



#### Explanation

From the ANOVA table, the calculated  $F$ -statistic is (mean square regression / mean square error) =  $(83.80 / 28.88) = 2.9017$ . From the  $F$  distribution table (4 df numerator, 21 df denominator) the critical  $F$  value is 2.84. Because 2.9017 is greater than 2.84, Williams rejects the null hypothesis and concludes that at least one of the independent variables has explanatory power.

(Module 1.2, LOS 1.e)

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**Question #58 - 60 of 139**

Question ID: 1479877

With respect to testing the validity of the model's results, Williams may wish to perform:

- A) both a Breusch-Godfrey test and a Breusch-Pagan test.
- B) a Breusch-Pagan test, but not Breusch-Godfrey.
- C) a Breusch-Godfrey test, but not a Breusch-Pagan test.

**Explanation**

Since the model utilized is not an autoregressive time series, a test for serial correlation is appropriate so the Breusch-Godfrey test would be used. The Breusch-Pagan test for heteroskedasticity would also be a good idea.

(Module 1.2, LOS 1.e)

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**Question #59 - 60 of 139**

Question ID: 1471974

When Williams ran the model, the computer said the  $R^2$  is 0.233. She examines the other output and concludes that this is the:

- A) adjusted  $R^2$  value.
- B) neither the unadjusted nor adjusted  $R^2$  value, nor the coefficient of correlation.
- C) unadjusted  $R^2$  value.

**Explanation**

This can be answered by recognizing that the unadjusted R-square is  $(335.2 / 941.6) = 0.356$ . Thus, the reported value must be the adjusted  $R^2$ . To verify this we see that the adjusted R-squared is:  $1 - ((26 - 1) / (26 - 4 - 1)) \times (1 - 0.356) = 0.233$ . Note that whenever there is more than one independent variable, the adjusted  $R^2$  will always be less than  $R^2$ .

(Module 1.2, LOS 1.e)

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**Question #60 - 60 of 139**

Question ID: 1471975

In preparing and using this model, Williams has *least* likely relied on which of the following assumptions?

- A) The residuals are homoscedastic.
- B) There is a linear relationship between the independent variables.



C) The disturbance or error term is normally distributed.



### Explanation

Multiple regression models assume that there is no linear relationship between two or more of the independent variables. The other answer choices are both assumptions of multiple regression.

(Module 1.2, LOS 1.e)

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### Question #61 of 139

Question ID: 1586007

Which of the following conditions will *least likely* affect the statistical inference about regression parameters by itself?

A) Unconditional heteroskedasticity.



B) Model misspecification.



C) Multicollinearity.



### Explanation

Unconditional heteroskedasticity does not impact the statistical inference concerning the parameters. Misspecified models have inconsistent and biased regression parameters. Multicollinearity results in unreliable estimates of regression parameters.

(Module 1.3, LOS 1.h)

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### Question #62 of 139

Question ID: 1472074

The management of a large restaurant chain believes that revenue growth is dependent upon the month of the year. Using a standard 12 month calendar, how many dummy variables must be used in a regression model that will test whether revenue growth differs by month?

A) 13.



B) 12.



C) 11.



### Explanation

The appropriate number of dummy variables is one less than the number of categories because the intercept captures the effect of the other effect. With 12 categories (months) the appropriate number of dummy variables is  $11 = 12 - 1$ . If the number of dummy variables equals the number of categories, it is possible to state any one of the independent dummy variables in terms of the others. This is a violation of the assumption of the multiple linear regression model that none of the independent variables are linearly related.

(Module 1.4, LOS 1.I)

Damon Washburn, CFA, is currently enrolled as a part-time graduate student at State University. One of his recent assignments for his course on Quantitative Analysis is to perform a regression analysis utilizing the concepts covered during the semester. He must interpret the results of the regression as well as the test statistics. Washburn is confident in his ability to calculate the statistics because the class is allowed to use statistical software. However, he realizes that the interpretation of the statistics will be the true test of his knowledge of regression analysis. His professor has given to the students a list of questions that must be answered by the results of the analysis.

Washburn has estimated a regression equation in which 160 quarterly returns on the S&P 500 are explained by three macroeconomic variables: employment growth (EMP) as measured by nonfarm payrolls, gross domestic product (GDP) growth, and private investment (INV). The results of the regression analysis are as follows:

| Coefficient Estimates |             |                               |
|-----------------------|-------------|-------------------------------|
| Parameter             | Coefficient | Standard Error of Coefficient |
| Intercept             | 9.50        | 3.40                          |
| EMP                   | -4.50       | 1.25                          |
| GDP                   | 4.20        | 0.76                          |
| INV                   | -0.30       | 0.16                          |

Other Data:

- Regression sum of squares (RSS) = 126.00
- Sum of squared errors (SSE) = 267.00
- BG-stat: Lag 1: 3.15; Lag 2: 3.22

| Degree of Freedom Denominator | Degree of Freedom Numerator |
|-------------------------------|-----------------------------|
|                               |                             |

|     | 1    | 2    | 3    |
|-----|------|------|------|
| 153 | 3.90 | 3.06 | 2.66 |
| 154 | 3.90 | 3.05 | 2.66 |
| 155 | 3.90 | 3.05 | 2.66 |
| 156 | 3.90 | 3.05 | 2.66 |
| 157 | 3.90 | 3.05 | 2.66 |
| 158 | 3.90 | 3.05 | 2.66 |

### Question #63 - 66 of 139

Question ID: 1471895

The percentage of the total variation in quarterly stock returns explained by the independent variables is *closest* to:

- A) 32%.
- B) 47%.
- C) 42%.



#### Explanation

The  $R^2$  is the percentage of variation in the dependent variable explained by the independent variables. The  $R^2$  is equal to the  $SS_{\text{Regression}}/SS_{\text{Total}}$ , where the  $SS_{\text{Total}}$  is equal to  $SS_{\text{Regression}} + SS_{\text{Error}}$ .  $R^2 = 126.00 / (126.00 + 267.00) = 32\%$ .

(Module 1.1, LOS 1.b)

### Question #64 - 66 of 139

Question ID: 1586009

Using a 5% level of significance, there is:

- A)** no evidence of serial correlation in the residuals. 
- B)** evidence of first-lag serial correlation in residuals. 
- C)** evidence of second-lag serial correlation in residuals. 

#### Explanation

$n=160$ ,  $k=3$ . For lag 1,  $p = 1$ . ( $n-k-p-1 = 155$ ). Critical F-stat (1, 155) = 3.90. BG Stat for lag 1 is given as 3.15. We fail to reject the null of no serial correlation in residuals. For lag 2, critical F (2, 154) = 3.05. BG stat for lag 2 is given as 3.22 and exceeds the critical value and hence we reject the null of no serial correlation at the 2nd lag.

(Module 1.3, LOS 1.i)

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### Question #65 - 66 of 139

Question ID: 1471897

What is the predicted quarterly stock return, given the following forecasts?

- Employment growth = 2.0%
- GDP growth = 1.0%
- Private investment growth = -1.0%

- A)** 4.4%. 
- B)** 4.7%. 
- C)** 5.0%. 

#### Explanation

Predicted quarterly stock return is  $9.50\% + (-4.50)(2.0\%) + (4.20)(1.0\%) + (-0.30)(-1.0\%) = 5.0\%$ .

(Module 1.1, LOS 1.b)

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### Question #66 - 66 of 139

Question ID: 1586010

Assuming a restricted model with all three variables removed and a 5% level of significance, the most appropriate conclusion is:

- A)** With an F-statistic of 0.472, we fail to reject the null hypothesis of all coefficients equal to zero. 
- B)** With an F-statistic of 2.66, we fail to reject the null hypothesis of all slope coefficients equal to zero. 



- C) With an F-statistic of 24.54, we reject the null hypothesis that all the slope coefficients are equal to zero.



### Explanation

We are testing that all slope coefficients are equal to 0.

$$F = [RSS/K] / [SSE/(n-k-1)] = (126/3) / (267/156) = 42/1.711 = 24.54$$

$$\text{Critical } F(3,156) = 2.66$$

F-stat > critical F > reject null that all slope coefficients are equal to zero.

(Module 1.2, LOS 1.e)

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### Question #67 of 139

Question ID: 1586005

Consider the following estimated regression equation:

$$\text{Sales}_i = 10.0 + 1.25 \text{ R\&D}_i + 1.0 \text{ ADV}_i - 2.0 \text{ COMP}_i + 8.0 \text{ CAP}_i$$

Sales are in millions of dollars. An analyst is given the following predictions on the independent variables: R&D = 5, ADV = 4, COMP = 10, and CAP = 40.

The predicted level of sales is *closest* to:

- A) \$320.25 million.
- B) \$300.25 million.
- C) \$310.25 million.



### Explanation

Predicted sales

$$= \$10 + 1.25 (5) + 1.0 (4) - 2.0 (10) + 8 (40)$$

$$= 10 + 6.25 + 4 - 20 + 320 = \$320.25$$

(Module 1.2, LOS 1.f)

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Miles Mason, CFA, works for ABC Capital, a large money management company based in New York. Mason has several years of experience as a financial analyst, but is currently working in the marketing department developing materials to be used by ABC's sales team for both existing and prospective clients. ABC Capital's client base consists primarily of large net worth individuals and Fortune 500 companies. ABC invests its clients' money in both publicly traded mutual funds as well as its own investment funds that are managed in-

house. Five years ago, roughly half of its assets under management were invested in the publicly traded mutual funds, with the remaining half in the funds managed by ABC's investment team. Currently, approximately 75% of ABC's assets under management are invested in publicly traded funds, with the remaining 25% being distributed among ABC's private funds. The managing partners at ABC would like to shift more of its client's assets away from publicly traded funds into ABC's proprietary funds, ultimately returning to a 50/50 split of assets between publicly traded funds and ABC funds. There are three key reasons for this shift in the firm's asset base. First, ABC's in-house funds have outperformed other funds consistently for the past five years. Second, ABC can offer its clients a reduced fee structure on funds managed in-house relative to other publicly traded funds. Lastly, ABC has recently hired a top fund manager away from a competing investment company and would like to increase his assets under management.

ABC Capital's upper management requested that current clients be surveyed in order to determine the cause of the shift of assets away from ABC funds. Results of the survey indicated that clients feel there is a lack of information regarding ABC's funds. Clients would like to see extensive information about ABC's past performance, as well as a sensitivity analysis showing how the funds will perform in varying market scenarios. Mason is part of a team that has been charged by upper management to create a marketing program to present to both current and potential clients of ABC. He needs to be able to demonstrate a history of strong performance for the ABC funds, and, while not promising any measure of future performance, project possible return scenarios. He decides to conduct a regression analysis on all of ABC's in-house funds. He is going to use 12 independent economic variables in order to predict each particular fund's return. Mason is very aware of the many factors that could minimize the effectiveness of his regression model, and if any are present, he knows he must determine if any corrective actions are necessary. Mason is using a sample size of 121 monthly returns.

### Question #68 - 70 of 139

Question ID: 1508634

Which of the following tests is *least likely* to be used to detect autocorrelation?

- A) Durbin-Watson.
- B) Breusch-Godfrey.
- C) Breusch-Pagan.



**Explanation**

Durbin-Watson and Breusch-Godfrey test statistic are used to detect autocorrelation. The Breusch-Pagan test is used to detect heteroskedasticity.

(Module 1.3, LOS 1.i)

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### Question #69 - 70 of 139

Question ID: 1472023

One of the most popular ways to correct heteroskedasticity is to:

- A) improve the specification of the model.
- B) adjust the standard errors.
- C) use robust standard errors.



#### Explanation

Using generalized least squares and calculating robust standard errors are possible remedies for heteroskedasticity. Improving specifications remedies serial correlation. The standard error cannot be adjusted, only the coefficient of the standard errors.

(Module 1.3, LOS 1.h)

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### Question #70 - 70 of 139

Question ID: 1479933

If a regression equation shows that no individual t-tests are significant, but the F-statistic is significant, the regression probably exhibits:

- A) serial correlation.
- B) multicollinearity.
- C) heteroskedasticity.



#### Explanation

Common indicators of multicollinearity include: high correlation ( $>0.7$ ) between independent variables, no individual t-tests are significant but the F-statistic is, and signs on the coefficients that are opposite of what is expected.

(Module 1.3, LOS 1.j)

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Lynn Carter, CFA, is an analyst in the research department for Smith Brothers in New York. She follows several industries, as well as the top companies in each industry. She provides research materials for both the equity traders for Smith Brothers as well as their retail customers. She routinely performs regression analysis on those companies that she follows to identify any emerging trends that could affect investment decisions.

Due to recent layoffs at the company, there has been some consolidation in the research department. Two research analysts have been laid off, and their workload will now be distributed among the remaining four analysts. In addition to her current workload, Carter will now be responsible for providing research on the airline industry. Pinnacle Airlines, a leader in the industry, represents a large holding in Smith Brothers' portfolio. Looking back over past research on Pinnacle, Carter recognizes that the company historically has been a strong performer in what is considered to be a very competitive industry. The stock price over the last 52-week period has outperformed that of other industry leaders, although Pinnacle's net income has remained flat. Carter wonders if the stock price of Pinnacle has become overvalued relative to its peer group in the market, and wants to determine if the timing is right for Smith Brothers to decrease its position in Pinnacle.

Carter decides to run a regression analysis, using the monthly returns of Pinnacle stock as the dependent variable and monthly returns of the airlines industry as the independent variable.

| Analysis of Variance Table (ANOVA) |                            |                        |                        |
|------------------------------------|----------------------------|------------------------|------------------------|
| Source                             | df<br>(Degrees of Freedom) | SS<br>(Sum of Squares) | Mean Square<br>(SS/df) |
| Regression                         | 1                          | 3,257 (RSS)            | 3,257 (MSR)            |
| Error                              | 8                          | 298 (SSE)              | 37.25 (MSE)            |
| Total                              | 9                          | 3,555 (SS Total)       |                        |

### Question #71 - 74 of 139

Question ID: 1479893

Which of the following is *least likely* to be an assumption regarding linear regression?

- A) A linear relationship exists between the dependent and independent variables. 
- B) The variance of the residuals is constant. 
- C) The independent variable is correlated with the residuals. 

**Explanation**

Although the linear regression model is fairly insensitive to minor deviations from any of these assumptions, the independent variable is typically uncorrelated with the residuals.

(Module 1.1, LOS 1.c)

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### Question #72 - 74 of 139

Question ID: 1471957

Based upon the information presented in the ANOVA table, what is the coefficient of determination?

- A) 0.084, indicating that the variability of industry returns explains about 8.4% of the variability of company returns. 
- B) 0.916, indicating that the variability of industry returns explains about 91.6% of the variability of company returns. 
- C) 0.839, indicating that company returns explain about 83.9% of the variability of industry returns. 

#### Explanation

The coefficient of determination ( $R^2$ ) is the percentage of the total variation in the dependent variable explained by the independent variable.

The  $R^2 = (RSS / SS) \text{ Total} = (3,257 / 3,555) = 0.916$ . This means that the variation of independent variable (the airline industry) explains 91.6% of the variations in the dependent variable (Pinnacle stock).

(Module 1.2, LOS 1.d)

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### Question #73 - 74 of 139

Question ID: 1479894

Based upon her analysis, Carter has derived the following regression equation:  $\hat{Y} = 1.75 + 3.25X_1$ . The predicted value of the Y variable equals 50.50, if the:

- A) predicted value of the independent variable equals 15. 
- B) predicted value of the dependent variable equals 15. 
- C) coefficient of the determination equals 15. 

#### Explanation

Note that the easiest way to answer this question is to plug numbers into the equation.

The predicted value for  $Y = 1.75 + 3.25(15) = 50.50$ .

The variable  $X_1$  represents the independent variable.

(Module 1.2, LOS 1.f)

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### Question #74 - 74 of 139

Question ID: 1479895

Carter realizes that although regression analysis is a useful tool when analyzing investments, there are certain limitations. Carter made a list of points describing limitations that Smith Brothers equity traders should be aware of when applying her research to their investment decisions.

- Point 1: Regression residuals may be homoskedastic.
- Point 2: Data from regression relationships tends to exhibit parameter instability.
- Point 3: Regression residuals may exhibit autocorrelation.
- Point 4: The variance of the error term may change with one or more independent variables.

When reviewing Carter's list, one of the Smith Brothers' equity traders points out that not all of the points describe regression analysis limitations. Which of Carter's points most accurately describes the limitations to regression analysis?

**A)** Points 1, 2, and 3.



**B)** Points 2, 3, and 4.



**C)** Points 1, 3, and 4.



#### Explanation

One of the basic assumptions of regression analysis is that the variance of the error terms is constant, or homoskedastic. Any violation of this assumption is called heteroskedasticity. Therefore, Point 1 is incorrect, but Point 4 is correct because it describes conditional heteroskedasticity, which results in unreliable estimates of standard errors. Points 2 and 3 also describe limitations of regression analysis.

(Module 1.1, LOS 1.c)

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### Question #75 of 139

Question ID: 1479903

When constructing a regression model to predict portfolio returns, an analyst runs a regression for the past five year period. After examining the results, she determines that an increase in interest rates two years ago had a significant impact on portfolio results for the time of the increase until the present. By performing a regression over two separate time periods, the analyst would be attempting to prevent which type of misspecification?

**A)** Incorrectly pooling data.



**B)** Inappropriate variable scaling.



**C)** Inappropriate variable form.



#### Explanation

The relationship between returns and the dependent variables can change over time, so it is critical that the data be pooled correctly. Running the regression for multiple sub-periods (in this case two) rather than one time period can produce more accurate results.

(Module 1.3, LOS 1.g)

#### Question #76 of 139

Question ID: 1471881

Which of the following statements *most* accurately interprets the following regression results at the given significance level?

| Variable  | p-value |
|-----------|---------|
| Intercept | 0.0201  |
| X1        | 0.0284  |
| X2        | 0.0310  |
| X3        | 0.0143  |

**A)** The variable X2 is statistically significantly different from zero at the 3% significance level.



**B)** The variables X1 and X2 are statistically significantly different from zero at the 2% significance level.



**C)** The variable X3 is statistically significantly different from zero at the 2% significance level.



#### Explanation

The  $p$ -value is the smallest level of significance for which the null hypothesis can be rejected. An independent variable is significant if the  $p$ -value is less than the stated significance level. In this example, X3 is the variable that has a  $p$ -value less than the stated significance level.

(Module 1.1, LOS 1.b)

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### Question #77 of 139

Question ID: 1479959

A high-yield bond analyst is trying to develop an equation using financial ratios to estimate the probability of a company defaulting on its bonds. A technique that can be used to develop this equation is:

- A) multiple linear regression adjusting for heteroskedasticity. 
- B) dummy variable regression. 
- C) logistic regression model. 

#### Explanation

The only one of the possible answers that estimates a probability of a discrete outcome is logit or logistic modeling.

(Module 1.4, LOS 1.m)

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### Question #78 of 139

Question ID: 1479867

Jason Fye, CFA, wants to check for seasonality in monthly stock returns (i.e., the January effect) after controlling for market cap and systematic risk. The type of model that Fye would *most* appropriately select is:

- A) Multiple regression model. 
- B) Neither multiple regression nor logistic regression. 
- C) logistic regression model. 

#### Explanation

Fye wants to test a theory of January effect on stock returns (dependent variable) using a dummy (January = 1, other months = 0), market cap, and beta (independent variables). A multiple regression model would be most appropriate. Because the dependent variable (stock returns) is not a qualitative variable, a logistic regression would not apply.

(Module 1.1, LOS 1.a)



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**Question #79 of 139**

Question ID: 1586011

An analyst is estimating whether company sales is related to three economic variables. The regression exhibits conditional heteroskedasticity, serial correlation, and multicollinearity. The analyst uses White and Newey-West corrected standard errors. Which of the following is *most* accurate?

- A) The regression will still exhibit multicollinearity, but the heteroskedasticity and serial correlation problems will be solved. 
- B) The regression will still exhibit serial correlation and multicollinearity, but the heteroskedasticity problem will be solved. 
- C) The regression will still exhibit heteroskedasticity and multicollinearity, but the serial correlation problem will be solved. 

**Explanation**

The correction mentioned solves for heteroskedasticity and serial correlation.

(Module 1.3, LOS 1.i)

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**Question #80 of 139**

Question ID: 1471868

Consider the following estimated regression equation, with the standard errors of the slope coefficients as noted:

$$\text{Sales}_i = 10.0 + 1.25 \text{ R\&D}_i + 1.0 \text{ ADV}_i - 2.0 \text{ COMP}_i + 8.0 \text{ CAP}_i$$

where the standard error for the estimated coefficient on R&D is 0.45, the standard error for the estimated coefficient on ADV is 2.2, the standard error for the estimated coefficient on COMP is 0.63, and the standard error for the estimated coefficient on CAP is 2.5.

The equation was estimated over 40 companies. Using a 5% level of significance, which of the estimated coefficients are significantly different from zero?

- A) R&D, ADV, COMP, and CAP. 
- B) R&D, COMP, and CAP only. 
- C) ADV and CAP only. 

**Explanation**

The critical t-values for  $40 - 4 - 1 = 35$  degrees of freedom and a 5% level of significance are  $\pm 2.03$ .

The calculated t-values are:

$$t \text{ for R\&D} = 1.25 / 0.45 = 2.777$$

$$t \text{ for ADV} = 1.0 / 2.2 = 0.455$$

$$t \text{ for COMP} = -2.0 / 0.63 = -3.175$$

$$t \text{ for CAP} = 8.0 / 2.5 = 3.2$$

Therefore, R&D, COMP, and CAP are statistically significant.

(Module 1.1, LOS 1.b)

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### Question #81 of 139

Question ID: 1471869

Consider the following regression equation:

$$\text{Sales}_i = 20.5 + 1.5 \text{ R\&D}_i + 2.5 \text{ ADV}_i - 3.0 \text{ COMP}_i$$

where Sales is dollar sales in millions, R&D is research and development expenditures in millions, ADV is dollar amount spent on advertising in millions, and COMP is the number of competitors in the industry.

Which of the following is NOT a correct interpretation of this regression information?

- A) If R&D and advertising expenditures are \$1 million each and there are 5 competitors, expected sales are \$9.5 million. 
- B) If a company spends \$1 more on R&D (holding everything else constant), sales are expected to increase by \$1.5 million. 
- C) One more competitor will mean \$3 million less in sales (holding everything else constant). 

#### Explanation

If a company spends \$1 *million* more on R&D (holding everything else constant), sales are expected to increase by \$1.5 million. Always be aware of the units of measure for the different variables.

(Module 1.1, LOS 1.b)

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### Question #82 of 139

Question ID: 1479914

Which of the following questions is *least likely* answered by using a qualitative dependent variable?

- A) Based on the following company-specific financial ratios, will company ABC enter bankruptcy? 
- B) Based on the following subsidiary and competition variables, will company XYZ divest itself of a subsidiary? 
- C) Based on the following executive-specific and company-specific variables, how many shares will be acquired through the exercise of executive stock options? 

#### Explanation

The number of shares can be a broad range of values and is, therefore, not considered a qualitative dependent variable.

(Module 1.3, LOS 1.h)

#### Question #83 of 139

Question ID: 1479874

Wilson estimated a regression that produced the following analysis of variance (ANOVA) table:

| Source     | Sum of squares | Degrees of freedom | Mean square |
|------------|----------------|--------------------|-------------|
| Regression | 100            | 1                  | 100.0       |
| Error      | 300            | 40                 | 7.5         |
| Total      | 400            | 41                 |             |

The values of  $R^2$  and the F-statistic to test the null hypothesis that slope coefficients on all variables are equal to zero are:

- A)  $R^2 = 0.25$  and  $F = 0.930$ . 
- B)  $R^2 = 0.20$  and  $F = 13.333$ . 
- C)  $R^2 = 0.25$  and  $F = 13.333$ . 

#### Explanation

$$R^2 = \text{RSS} / \text{SST} = 100 / 400 = 0.25$$

The F-statistic is equal to the ratio of the mean squared regression to the mean squared error.

$$F = 100 / 7.5 = 13.333$$

(Module 1.2, LOS 1.e)

### Question #84 of 139

Question ID: 1586003

May Jones estimated a regression that produced the following analysis of variance (ANOVA) table:

| Source     | Sum of squares | Degrees of freedom | Mean square |
|------------|----------------|--------------------|-------------|
| Regression | 20             | 1                  | 20          |
| Error      | 80             | 40                 | 2           |
| Total      | 100            | 41                 |             |

The values of  $R^2$  and the  $F$ -statistic for joint test of significance of all the slope coefficients are:

**A)**  $R^2 = 0.25$  and  $F = 0.909$ .



**B)**  $R^2 = 0.20$  and  $F = 10$ .



**C)**  $R^2 = 0.25$  and  $F = 10$ .



#### Explanation

$$R^2 = \text{RSS} / \text{SST} = 20 / 100 = 0.20$$

The  $F$ -statistic is equal to the ratio of the mean squared regression to the mean squared error.

$$F = 20 / 2 = 10$$

(Module 1.2, LOS 1.e)

Dave Turner is a security analyst who is using regression analysis to determine how well two factors explain returns for common stocks. The independent variables are the natural logarithm of the number of analysts following the companies,  $\text{Ln}(\text{no. of analysts})$ , and the natural logarithm of the market value of the companies,  $\text{Ln}(\text{market value})$ . The regression

output generated from a statistical program is given in the following tables. Each p-value corresponds to a two-tail test.

Turner plans to use the result in the analysis of two investments. WLK Corp. has twelve analysts following it and a market capitalization of \$2.33 billion. NGR Corp. has two analysts following it and a market capitalization of \$47 million.

**Table 1: Regression Output**

| Variable            | Coefficient | Standard Error of the Coefficient | t-statistic | p-value |
|---------------------|-------------|-----------------------------------|-------------|---------|
| Intercept           | 0.043       | 0.01159                           | 3.71        | < 0.001 |
| Ln(No. of Analysts) | -0.027      | 0.00466                           | -5.80       | < 0.001 |
| Ln(Market Value)    | 0.006       | 0.00271                           | 2.21        | 0.028   |

**Table 2: ANOVA**

|            | Degrees of Freedom | Sum of Squares | Mean Square |
|------------|--------------------|----------------|-------------|
| Regression | 2                  | 0.103          | 0.051       |
| Residual   | 194                | 0.559          | 0.003       |
| Total      | 196                | 0.662          |             |

### Question #85 - 88 of 139

Question ID: 1471887

If the number of analysts on NGR Corp. were to double to 4, the change in the forecast of NGR would be *closest* to?

**A)** -0.035.



**B)** -0.055.



**C)** -0.019.



#### Explanation

Initially, the estimate is  $0.1303 = 0.043 + \ln(2)(-0.027) + \ln(47000000)(0.006)$

Then, the estimate is  $0.1116 = 0.043 + \ln(4)(-0.027) + \ln(47000000)(0.006)$

$0.1116 - 0.1303 = -0.0187$ , or  $-0.019$

(Module 1.1, LOS 1.b)

**Question #86 - 88 of 139**

Question ID: 1585993

Based on a  $R^2$  calculated from the information in Table 2, the analyst should conclude that the number of analysts and  $\ln(\text{market value})$  of the firm explain:

- A) 18.4% of the variation in returns.
- B) 84.4% of the variation in returns.
- C) 15.6% of the variation in returns.

**Explanation**

$R^2$  is the percentage of the variation in the dependent variable (in this case, variation of returns) explained by the set of independent variables.  $R^2$  is calculated as follows:  $R^2 = (\text{SSR} / \text{SST}) = (0.103 / 0.662) = 15.6\%$ .

(Module 1.1, LOS 1.b)

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**Question #87 - 88 of 139**

Question ID: 1507766

What is the F-statistic for the hypothesis that all slope coefficients are not statistically significantly different from 0? And, what can be concluded from its value at a 1% level of significance?

- A)  $F = 5.80$ , reject a hypothesis that both of the slope coefficients are equal to zero.
- B)  $F = 17.00$ , reject a hypothesis that both of the slope coefficients are equal to zero.
- C)  $F = 1.97$ , fail to reject a hypothesis that both of the slope coefficients are equal to zero.

**Explanation**

The F-statistic is calculated as follows:  $F = \text{MSR} / \text{MSE} = 0.051 / 0.003 = 17.00$ ; and  $17.00 > 4.61$ , which is the critical F-value for the given degrees of freedom and a 1% level of significance. However, when F-values are in excess of 10 for a large sample like this, a table is not needed to know that the value is significant.

(Module 1.1, LOS 1.b)

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**Question #88 - 88 of 139**

Question ID: 1585994

Upon further analysis, Turner concludes that multicollinearity is a problem. What might have prompted this further analysis and what is intuition behind the conclusion?

At least one of the  $t$ -statistics was not significant, the F-statistic was significant,  
**A)** and a positive relationship between the number of analysts and the size of the firm would be expected. 

At least one of the  $t$ -statistics was not significant, the F-statistic was not  
**B)** significant, and a positive relationship between the number of analysts and the size of the firm would be expected. 

At least one of the  $t$ -statistics was not significant, the F-statistic was significant,  
**C)** and an intercept not significantly different from zero would be expected. 

### Explanation

Multicollinearity occurs when there is a high correlation among independent variables and may exist if there is a significant F-statistic for the fit of the regression model, but at least one insignificant independent variable when we expect all of them to be significant. In this case the coefficient on  $\ln(\text{market value})$  was not significant at the 1% level, but the F-statistic was significant. It would make sense that the size of the firm, i.e., the market value, and the number of analysts would be positively correlated.

(Module 1.1, LOS 1.b)

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### Question #89 of 139

Question ID: 1479934

A regression with three independent variables have VIF values of 3, 4, and 2 for the first, second, and third independent variables, respectively. Which of the following conclusions is *most* appropriate?

**A)** Only variable two has a problem with multicollinearity. 

**B)** Multicollinearity does not seem to be a problem with the model. 

**C)** Total VIF of 9 indicates a serious multicollinearity problem. 

### Explanation

Multicollinearity occurs when an independent variable is highly correlated with a linear combination of the remaining independent variables. VIF values exceeding 5 need to be investigated while values exceeding 10 indicate strong evidence of multicollinearity.

(Module 1.3, LOS 1.j)

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Henry Hilton, CFA, is undertaking an analysis of the bicycle industry. He hypothesizes that bicycle sales (SALES) are a function of three factors: the population under 20 (POP), the level of disposable income (INCOME), and the number of dollars spent on advertising (ADV). All data are measured in millions of units. Hilton gathers data for the last 20 years. Which of the follow regression equations *correctly* represents Hilton's hypothesis?

A)  $SALES = \alpha \times \beta_1 POP \times \beta_2 INCOME \times \beta_3 ADV \times \epsilon$



B)  $INCOME = \alpha + \beta_1 POP + \beta_2 SALES + \beta_3 ADV + \epsilon$



C)  $SALES = \alpha + \beta_1 POP + \beta_2 INCOME + \beta_3 ADV + \epsilon$



### Explanation

SALES is the dependent variable. POP, INCOME, and ADV should be the independent variables (on the right hand side) of the equation (in any order). Regression equations are additive.

(Module 1.1, LOS 1.b)

Jessica Jenkins, CFA, is looking at the retail property sector for her manager. She is undertaking a top down review as she feels this is the best way to analyze the industry segment. To predict U.S. property starts (housing), she has used regression analysis.

Jessica included the following variables in her analysis:

- Average nominal interest rates during each year (as a decimal)
- Annual GDP per capita in \$'000

Given these variables, the following output was generated from 30 years of data:

### Exhibit 1 – Results from regressing housing starts (in millions) on interest rates and GDP per capita

|                |  | Coefficient | Standard Error | T-statistic |
|----------------|--|-------------|----------------|-------------|
| Intercept      |  | 0.42        |                | 3.1         |
| Interest rate  |  | -1.0        |                | -2.0        |
| GDP per capita |  | 0.03        |                | 0.7         |



| <b>ANOVA</b>  | <b>df</b> | <b>SS</b> | <b>MSS</b> | <b>F</b> |
|---------------|-----------|-----------|------------|----------|
| Regression    | 2         | 3.896     | 1.948      | 21.644   |
| Residual      | 27        | 2.431     | 0.090      |          |
| Total         | 29        | 6.327     |            |          |
| Observations  | 30        |           |            |          |
| Durbin-Watson | 1.27      |           |            |          |

**Exhibit 2 - Critical Values for F-Distribution at 5% Level of Significance**

| <b>Degrees of Freedom for the Denominator</b> | <b>Degrees of Freedom (df) for the Numerator</b> |          |          |
|-----------------------------------------------|--------------------------------------------------|----------|----------|
|                                               | <b>1</b>                                         | <b>2</b> | <b>3</b> |
| 26                                            | 4.23                                             | 3.37     | 2.98     |
| 27                                            | 4.21                                             | 3.35     | 2.96     |
| 28                                            | 4.20                                             | 3.34     | 2.95     |
| 29                                            | 4.18                                             | 3.33     | 2.93     |
| 30                                            | 4.17                                             | 3.32     | 2.92     |
| 31                                            | 4.16                                             | 3.31     | 2.91     |
| 32                                            | 4.15                                             | 3.30     | 2.90     |

The following variable estimates have been made for 20X7:

GDP per capita = \$46,700

Interest rate = 7%

### Question #91 - 93 of 139

Question ID: 1479880

Using the regression model represented in Exhibit 1, what is the predicted number of housing starts for 20X7?

A) 1,751,000.



B) 1,394.



C) 1,394,420.



#### Explanation

Housing starts =  $0.42 - (1 \times 0.07) + (0.03 \times 46.7) = 1.751$  million

(Module 1.2, LOS 1.e)

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### Question #92 - 93 of 139

Question ID: 1472031

Which of the following statements *best* describes the explanatory power of the estimated regression?

A) The residual standard error of only 0.3 indicates that the regression equation is a good fit for the sample data.



B) The independent variables explain 61.58% of the variation in housing starts.



C) The large *F*-statistic indicates that both independent variables help explain changes in housing starts.



#### Explanation

The coefficient of determination is the statistic used to identify explanatory power. This can be calculated from the ANOVA table as  $3.896 / 6.327 \times 100 = 61.58\%$ .

The residual standard error of 0.3 indicates that the standard deviation of the residuals is 0.3 million housing starts. Without knowledge of the data for the dependent variable, it is not possible to assess whether this is a small or a large error.

The *F*-statistic does not enable us to conclude on *both* independent variables. It only allows us to reject the hypothesis that all regression coefficients are zero and accept the hypothesis that at *least one* isn't.

(Module 1.2, LOS 1.d)

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### Question #93 - 93 of 139

Question ID: 1479882

Which of the following is the *least* appropriate statement in relation to *R*-square and adjusted *R*-square?

- A) Adjusted *R*-square can be higher than the coefficient of determination for a model with a good fit. 
- B) Adjusted *R*-square decreases when the added independent variable adds little value to the regression model. 
- C) *R*-square typically increases when new independent variables are added to the regression. 

#### Explanation

Adjusted *R*-squared cannot exceed *R*-squared (or coefficient of determination) for a multiple regression. .

(Module 1.2, LOS 1.d)

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### Question #94 of 139

Question ID: 1472073

A fund has changed managers twice during the past 10 years. An analyst wishes to measure whether either of the changes in managers has had an impact on performance. *R* is the return on the fund, and *M* is the return on a market index. Which of the following regression equations can appropriately measure the desired impacts?

- A) The desired impact cannot be measured. 

$R = a + bM + c_1D_1 + c_2D_2 + c_3D_3 + \varepsilon$ , where  $D_1 = 1$  if the return is from the first

**B)** manager, and  $D_2 = 1$  if the return is from the second manager, and  $D_3 = 1$  if the return is from the third manager. ❌

$R = a + bM + c_1D_1 + c_2D_2 + \varepsilon$ , where  $D_1 = 1$  if the return is from the first

**C)** manager, and  $D_2 = 1$  if the return is from the third manager. ✅

### Explanation

The effect needs to be measured by two distinct dummy variables. The use of three variables will cause collinearity, and the use of one dummy variable will not appropriately specify the manager impact.

(Module 1.4, LOS 1.I)

Raul Gloucester, CFA, is analyzing the returns of a fund that his company offers. He tests the fund's sensitivity to a small capitalization index and a large capitalization index, as well as to whether the January effect plays a role in the fund's performance. He uses two years of monthly returns data, and runs a regression of the fund's return on the indexes and a January-effect qualitative variable. The "January" variable is 1 for the month of January and zero for all other months. The results of the regression are shown in the tables below.

| Regression Statistics   |          |
|-------------------------|----------|
| Multiple R              | 0.817088 |
| R <sup>2</sup>          | 0.667632 |
| Adjusted R <sup>2</sup> | 0.617777 |
| Standard Error          | 1.655891 |
| Observations            | 24       |

| ANOVA      |    |          |          |
|------------|----|----------|----------|
|            | df | SS       | MS       |
| Regression | 3  | 110.1568 | 36.71895 |
| Residual   | 20 | 54.8395  | 2.741975 |
| Total      | 23 | 164.9963 |          |

|           | Coefficients | Standard Error | t-Statistic |
|-----------|--------------|----------------|-------------|
| Intercept | -0.23821     | 0.388717       | -0.61282    |

|                 |          |          |          |
|-----------------|----------|----------|----------|
| January         | 2.560552 | 1.232634 | 2.077301 |
| Small Cap Index | 0.231349 | 0.123007 | 1.880778 |
| Large Cap Index | 0.951515 | 0.254528 | 3.738359 |

**Exhibit 1: Partial F-Table (5% Level of Significance)**

| Degree of Freedom Denominator | Degree of Freedom Numerator |      |      |
|-------------------------------|-----------------------------|------|------|
|                               | 1                           | 2    | 3    |
| 18                            | 4.41                        | 3.55 | 3.16 |
| 19                            | 4.38                        | 3.52 | 3.13 |
| 20                            | 4.35                        | 3.49 | 3.10 |
| 21                            | 4.32                        | 3.47 | 3.07 |
| 22                            | 4.30                        | 3.44 | 3.05 |
| 23                            | 4.28                        | 3.42 | 3.03 |

Gloucester plans to test for serial correlation and conditional and unconditional heteroskedasticity.

### Question #95 - 100 of 139

Question ID: 1479925

The percent of the variation in the fund's return that is explained by the regression is:

**A)** 61.78%.



**B)** 81.71%.



**C)** 66.76%.



#### Explanation

The  $R^2$  tells us how much of the change in the dependent variable is explained by the changes in the independent variables in the regression: 0.667632.

(Module 1.2, LOS 1.d)

### Question #96 - 100 of 139

Question ID: 1479926

Suppose the Breusch-Godfrey statistic is 3.22. At a 5% level of significance, which of the following is the *most* accurate conclusion regarding the presence of serial correlation (at two lags) in the residuals?

- A) No, because the BG statistic is less than the critical test statistic of 3.55, we don't have evidence of serial correlation. 
- B) Yes, because the BG statistic exceeds the critical test statistic of 3.16, there is evidence of serial correlation. 
- C) No, because the BG statistic is less than the critical test statistic of 3.49, we don't have evidence of serial correlation. 

### Explanation

Number of lags tested =  $p = 2$ . The appropriate test statistic for BG test is  $F$ -stat with ( $p = 2$ ) and ( $n - p - k - 1 = 18$ ) degrees of freedom. From the table, critical value = 3.55.

(Module 1.3, LOS 1.i)

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## Question #97 - 100 of 139

Question ID: 1479927

Gloucester subsequently revises the model to exclude the small cap index and finds that the revised model has a RSS of 106.332. Which of the following statements is *most* accurate? At a 5% level of significance, the test statistic:

- A) of 13.39 indicates that we cannot reject the hypothesis that the coefficient of small-cap index is significantly different from 0. 
- B) of 1.30 indicates that we cannot reject the hypothesis that the coefficient of small-cap index is not significantly different from 0. 
- C) of 4.35 indicates that we cannot reject the hypothesis that the coefficient of small-cap index is significantly different from 0. 

### Explanation

$$SSE_R = SST - RSS_R = 164.9963 - 106.3320 = 58.6643$$

$$F = [(SSE_R - SSE_U) / q] / [SSE_U / (n - k - 1)] = [(58.6643 - 54.8395) / 1] / (54.8395 / 20) = 3.8248 / 2.742 = 1.30$$

$$\text{Critical } F_{(1, 20)} = 4.35 \text{ (from Exhibit 1)}$$

Since the test statistic is not greater than the critical value, we cannot reject the null hypothesis that  $b_2 = 0$ .

(Module 1.2, LOS 1.e)

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### Question #98 - 100 of 139

Question ID: 1586013

The *best* test for unconditional heteroskedasticity is:

- A) the Breusch-Godfrey test only. 
- B) the Breusch-Pagan test only. 
- C) neither the Durbin-Watson test nor the Breusch-Pagan test. 

#### Explanation

Breusch-Godfrey and Durbin-Watson tests are for serial correlation. The Breusch-Pagan test is for *conditional* heteroskedasticity; it tests to see if the size of the independent variables influences the size of the residuals. Although tests for unconditional heteroskedasticity exist, they are not part of the CFA curriculum, and unconditional heteroskedasticity is generally considered less serious than conditional heteroskedasticity.

(Module 1.3, LOS 1.i)

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### Question #99 - 100 of 139

Question ID: 1479929

In the month of January, if both the small and large capitalization index have a zero return, we would expect the fund to have a return equal to:

- A) 2.799. 
- B) 2.561. 
- C) 2.322. 

#### Explanation

The forecast of the return of the fund would be the intercept plus the coefficient on the January effect:  $2.322 = -0.238214 + 2.560552$ .

(Module 1.2, LOS 1.f)

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### Question #100 - 100 of 139

Question ID: 1479930

Assuming (for this question only) that the *F*-test was significant but that the *t*-tests of the independent variables were insignificant, this would *most likely* suggest:

- A) conditional heteroskedasticity. 

B) serial correlation.



C) multicollinearity.



### Explanation

When the *F*-test and the *t*-tests conflict, multicollinearity is indicated.

(Module 1.3, LOS 1.j)

Quin Tan Liu, CFA, is looking at the retail property sector for her manager. She is undertaking a top down review as she feels this is the best way to analyze the industry segment. To predict U.S. property starts (housing), she has used regression analysis.

Liu included the following variables in her analysis:

- Average nominal interest rates during each year (as a decimal)
- Annual GDP per capita in \$'000

Given these variables the following output was generated from 30 years of data:

### Exhibit 1 – Results from Regressing Housing Starts (in Millions) on Interest Rates and GDP Per Capita

|                |           | Coefficient | Standard Error | T-statistic |
|----------------|-----------|-------------|----------------|-------------|
| Intercept      |           | 0.42        |                | 3.1         |
| Interest rate  |           | −1.0        |                | −2.0        |
| GDP per capita |           | 0.03        |                | 0.7         |
| <b>ANOVA</b>   | <b>df</b> | <b>SS</b>   | <b>MSS</b>     | <b>F</b>    |
| Regression     | 2         | 3.896       | 1.948          | 21.644      |
| Residual       | 27        | 2.431       | 0.090          |             |
| Total          | 29        | 6.327       |                |             |
| Observations   | 30        |             |                |             |
| Durbin-Watson  | 1.22      |             |                |             |

### Exhibit 2 - Critical Values for F-Distribution at 5% Level of Significance

| Degrees of Freedom for the Denominator | Degrees of Freedom (df) for the Numerator |      |      |
|----------------------------------------|-------------------------------------------|------|------|
|                                        | 1                                         | 2    | 3    |
| 26                                     | 4.23                                      | 3.37 | 2.98 |



|    |      |      |      |
|----|------|------|------|
| 27 | 4.21 | 3.35 | 2.96 |
| 28 | 4.20 | 3.34 | 2.95 |
| 29 | 4.18 | 3.33 | 2.93 |
| 30 | 4.17 | 3.32 | 2.92 |
| 31 | 4.16 | 3.31 | 2.91 |
| 32 | 4.15 | 3.30 | 2.90 |

The following variable estimates have been made for 20X7:

GDP per capita = \$46,700

Interest rate = 7%

### Question #101 - 103 of 139

Question ID: 1508632

Using the regression model represented in Exhibit 1, what is the predicted number of housing starts for 20X7?

- A) 1,751,000.
- B) 1,394,420.
- C) 1,394.



#### Explanation

Housing starts =  $0.42 - (1 \times 0.07) + (0.03 \times 46.7) = 1.751$  million

(Module 1.2, LOS 1.f)

### Question #102 - 103 of 139

Question ID: 1479886

Which of the following statements *best* describes the explanatory power of the estimated regression?

- A) The large *F*-statistic indicates that both independent variables help explain changes in housing starts.
- B) The independent variables explain 61.58% of the variation in housing starts.
- C) The residual standard error of only 0.3 indicates that the regression equation is a good fit for the sample data.



### Explanation

The coefficient of determination is the statistic used to identify explanatory power. This can be calculated from the ANOVA table as  $3.896/6.327 \times 100 = 61.58\%$ .

The residual standard error of 0.3 indicates that the standard deviation of the residuals is 0.3 million housing starts. Without knowledge of the data for the dependent variable it is not possible to assess whether this is a small or a large error.

The *F*-statistic does not enable us to conclude on both independent variables. It only allows us to reject the hypothesis that all regression coefficients are zero and accept the hypothesis that at least one isn't.

(Module 1.2, LOS 1.d)

(Module 1.2, LOS 1.e)

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### Question #103 - 103 of 139

Question ID: 1543892

Which of the following is the *least appropriate* statement in relation to R-square and adjusted R-square:

- A) Adjusted R-square decreases when the added independent variable adds little value to the regression model. 
- B) R-square typically increases when new independent variables are added to the regression regardless of their explanatory power. 
- C) Adjusted R-square is a value between 0 and 1 and can be interpreted as a percentage. 

### Explanation

Adjusted R-square can be negative for a large number of independent variables that have no explanatory power. The other two statements are correct.

(Module 1.2, LOS 1.d)

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### Question #104 of 139

Question ID: 1479919

Which of the following is *least likely* a method of detecting serial correlations?

- A) A scatter plot of the residuals over time. 
- B) The Breusch-Pagan test. 
- C) The Breusch-Godfrey test. 

### Explanation

The Breusch-Pagan test is a test of the heteroskedasticity and not of serial correlation.

(Module 1.3, LOS 1.i)

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### Question #105 of 139

Question ID: 1471882

When interpreting the results of a multiple regression analysis, which of the following terms represents the value of the dependent variable when the independent variables are all equal to zero?

A) Slope coefficient.



B) Intercept term.



C)  $p$ -value.



### Explanation

The intercept term is the value of the dependent variable when the independent variables are set to zero.

(Module 1.1, LOS 1.b)

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### Question #106 of 139

Question ID: 1471870

Consider the following regression equation:

$$\text{Sales}_i = 10.0 + 1.25 \text{ R\&D}_i + 1.0 \text{ ADV}_i - 2.0 \text{ COMP}_i + 8.0 \text{ CAP}_i$$

where Sales is dollar sales in millions, R&D is research and development expenditures in millions, ADV is dollar amount spent on advertising in millions, COMP is the number of competitors in the industry, and CAP is the capital expenditures for the period in millions of dollars.

Which of the following is NOT a correct interpretation of this regression information?

A) If a company spends \$1 million more on capital expenditures (holding everything else constant), Sales are expected to increase by \$8.0 million.



B) One more competitor will mean \$2 million less in Sales (holding everything else constant).



If R&D and advertising expenditures are \$1 million each, there are 5  
**C)** competitors, and capital expenditures are \$2 million, expected Sales are \$8.25 million. 

**Explanation**

Predicted sales = \$10 + 1.25 + 1 – 10 + 16 = \$18.25 million.

(Module 1.1, LOS 1.b)

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Autumn Voiku is attempting to forecast sales for Brookfield Farms based on a multiple regression model. Voiku has constructed the following model:

$$\text{sales} = b_0 + (b_1 \times \text{CPI}) + (b_2 \times \text{IP}) + (b_3 \times \text{GDP}) + \varepsilon_t$$

Where:

sales = \$ change in sales (in 000's)

CPI = change in the consumer price index

IP = change in industrial production (millions)

GDP = change in GDP (millions)

All changes in variables are in percentage terms.

Voiku uses monthly data from the previous 180 months of sales data and for the independent variables. The model estimates (with coefficient standard errors in parentheses) are:

|         |       |               |              |                |
|---------|-------|---------------|--------------|----------------|
| SALES = | 10.2  | + (4.6 × CPI) | + (5.2 × IP) | + (11.7 × GDP) |
| p-value | 0.001 | 0.17          | 0.11         | 0.09           |

The sum of squared errors is 140.3 and the total sum of squares is 368.7.

Voiku calculates the unadjusted  $R^2$ , the adjusted  $R^2$ , and the standard error of estimate to be 0.592, 0.597, and 0.910, respectively.

Voiku is concerned that one or more of the assumptions underlying multiple regression has been violated in her analysis. In a conversation with Dave Grimble, CFA, a colleague who is considered by many in the firm to be a quant specialist, Voiku says, "It is my understanding that there are five assumptions of a multiple regression model:"

|               |                                                                                 |
|---------------|---------------------------------------------------------------------------------|
| Assumption 1: | There is a linear relationship between the dependent and independent variables. |
|---------------|---------------------------------------------------------------------------------|

|               |                                                                                                                       |
|---------------|-----------------------------------------------------------------------------------------------------------------------|
| Assumption 2: | The independent variables are not random, and there is zero correlation between any two of the independent variables. |
| Assumption 3: | The residual term is normally distributed with an expected value of zero.                                             |
| Assumption 4: | The residuals are serially correlated.                                                                                |
| Assumption 5: | The variance of the residuals is constant.                                                                            |

Grimbles agrees with Miller's assessment of the assumptions of multiple regression.

Voiku tests and fails to reject each of the following four null hypotheses at the 99% confidence interval:

|               |                                                                                    |
|---------------|------------------------------------------------------------------------------------|
| Hypothesis 1: | The coefficient on GDP is negative.                                                |
| Hypothesis 2: | The intercept term is equal to -4.                                                 |
| Hypothesis 3: | A 2.6% increase in the CPI will result in an increase in sales of more than 12.0%. |
| Hypothesis 4: | A 1% increase in industrial production will result in a 1% decrease in sales.      |

Figure 1: Partial F-Table critical values for right-hand tail area equal to 0.05

|           | df1 = 1 | df1 = 3 | df1 = 5 |
|-----------|---------|---------|---------|
| df2 = 170 | 3.90    | 2.66    | 2.27    |
| df2 = 176 | 3.89    | 2.66    | 2.27    |
| df2 = 180 | 3.89    | 2.65    | 2.26    |

### Question #107 - 109 of 139

Question ID: 1471907

Concerning the assumptions of multiple regression, Grimbles is:

- A) incorrect to agree with Voiku's list of assumptions because one of the assumptions is stated incorrectly. 
- B) incorrect to agree with Voiku's list of assumptions because two of the assumptions are stated incorrectly. 
- C) correct to agree with Voiku's list of assumptions. 

**Explanation**

Assumption 2 is stated incorrectly. Some correlation between independent variables is unavoidable; and high correlation results in multicollinearity. However, an exact linear relationship between linear combinations of two or more independent variables should not exist.

Assumption 4 is also stated incorrectly. The assumption is that the residuals are serially uncorrelated (i.e., they are not serially correlated).

(Module 1.1, LOS 1.b)

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### Question #108 - 109 of 139

Question ID: 1471909

The *most* appropriate decision with regard to the F-statistic for testing the null hypothesis that all of the independent variables are simultaneously equal to zero at the 5 percent significance level is to:

- A) reject the null hypothesis because the F-statistic is larger than the critical F-value of 2.66. 
- B) fail to reject the null hypothesis because the F-statistic is smaller than the critical F-value of 2.66. 
- C) reject the null hypothesis because the F-statistic is larger than the critical F-value of 3.19. 

#### Explanation

$RSS = 368.7 - 140.3 = 228.4$ ,  $F\text{-statistic} = (228.4 / 3) / (140.3 / 176) = 95.51$ . The critical value for a one-tailed 5% F-test with 3 and 176 degrees of freedom is 2.66. Because the F-statistic is greater than the critical F-value, the null hypothesis that all of the independent variables are simultaneously equal to zero should be rejected.

(Module 1.1, LOS 1.b)

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### Question #109 - 109 of 139

Question ID: 1585996

The multiple regression, as specified, *most likely* suffers from:

- A) serial correlation of the error terms. 
- B) heteroskedasticity. 
- C) multicollinearity. 

#### Explanation

The regression is highly significant (based on the F-stat in Part 3), but the individual coefficients are not (all p-values > 0.05). This is a result of a regression with significant multicollinearity problems.

(Module 1.1, LOS 1.b)

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### Question #110 of 139

Question ID: 1489310

Jacob Warner, CFA, is evaluating a regression analysis recently published in a trade journal that hypothesizes that the annual performance of the S&P 500 stock index can be explained by movements in the Federal Funds rate and the U.S. Producer Price Index (PPI). Which of the following statements regarding his analysis is *most accurate*?

- A) If the  $p$ -value of a variable is less than the significance level, the null hypothesis cannot be rejected. 
- B) If the  $t$ -value of a variable is less than the significance level, the null hypothesis should be rejected. 
- C) If the  $p$ -value of a variable is less than the significance level, the null hypothesis can be rejected. 

#### Explanation

The  $p$ -value is the smallest level of significance for which the null hypothesis can be rejected. Therefore, for any given variable, if the  $p$ -value of a variable is less than the significance level, the null hypothesis can be rejected and the variable is considered to be statistically significant.

(Module 1.1, LOS 1.b)

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A real estate agent wants to develop a model to predict the selling price of a home. The agent believes that the most important variables in determining the price of a house are its size (in square feet) and the number of bedrooms. Accordingly, he takes a random sample of 32 homes that has recently been sold. The results of the regression are:

|                    | Coefficient | Standard Error | t-statistics |
|--------------------|-------------|----------------|--------------|
| Intercept          | 66,500      | 59,292         | 1.12         |
| House Size         | 74.30       | 21.11          | 3.52         |
| Number of Bedrooms | 10306       | 3230           | 3.19         |

$$R^2 = 0.56; F = 40.73$$

Selected F- table values for significance level of 0.05:

|    | 1    | 2    |
|----|------|------|
| 28 | 4.20 | 3.34 |
| 29 | 4.18 | 3.33 |
| 30 | 4.17 | 3.32 |
| 32 | 4.15 | 3.29 |

(Degrees of freedom for the numerator in columns; Degrees of freedom for the denominator in rows)

Additional information regarding this multiple regression:

1. Variance of error is not constant across the 32 observations.
2. The two variables (size of the house and the number of bedrooms) are highly correlated.
3. The error variance is not correlated with the size of the house nor with the number of bedrooms.

### Question #111 - 113 of 139

Question ID: 1479936

The predicted price of a house that has 2,000 square feet of space and 4 bedrooms is *closest* to:

A) \$114,000.



B) \$256,000.



C) \$185,000.



#### Explanation

$$66,500 + 74.30(2,000) + 10,306(4) = \$256,324$$

(Module 1.2, LOS 1.f)

### Question #112 - 113 of 139

Question ID: 1479937

The conclusion from the hypothesis test of  $H_0: b_1 = b_2 = 0$ , is that the null hypothesis should:



- not be rejected as the calculated F of 40.73 is greater than the critical value of 3.29. 
- A)** 3.29. 
- B)** be rejected as the calculated F of 40.73 is greater than the critical value of 3.33. 
- C)** be rejected as the calculated F of 40.73 is greater than the critical value of 3.29. 

### Explanation

We can reject the null hypothesis that coefficients of both independent variables equal 0. The F value for comparison is  $F_{2,29} = 3.33$ . The degrees of freedom in the numerator is 2; equal to the number of independent variables. Degrees of freedom for the denominator is  $32 - (2+1) = 29$ . The critical value of the F-test needed to reject the null hypothesis is thus 3.33. The actual value of the F-test statistic is 40.73, so the null hypothesis should be rejected, as the calculated F of 40.73 is greater than the critical value of 3.33.

(Module 1.2, LOS 1.e)

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### Question #113 - 113 of 139

Question ID: 1479938

Which of the following is *most likely* to present a problem in using this regression for forecasting?

- A)** Heteroskedasticity. 
- B)** Multicollinearity. 
- C)** Autocorrelation. 

### Explanation

Multicollinearity is present in a regression model when some linear combination of the independent variables are highly correlated. We are told that the two independent variables in this question are highly correlated. We also recognize that unconditional heteroskedasticity is present – but this would not pose any major problems in using this model for forecasting. No information is given about autocorrelation in residuals, but this is generally a concern with time series data (in this case, the model uses cross-sectional data).

(Module 1.3, LOS 1.j)

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### Question #114 of 139

Question ID: 1472026

Which of the following statements regarding serial correlation that might be encountered in regression analysis is *least* accurate?

- A) Serial correlation occurs least often with time series data. 
- B) Serial correlation does not affect consistency of regression coefficients. 
- C) Positive serial correlation and heteroskedasticity can both lead to Type I errors. 

### Explanation

Serial correlation, which is sometimes referred to as autocorrelation, occurs when the residual terms are correlated with one another, and is most frequently encountered with time series data. Positive serial correlation can lead to standard errors that are too small, which will cause computed t-statistics to be larger than they should be, which will lead to too many Type I errors (i.e. the rejection of the null hypothesis when it is actually true). Serial correlation however does not affect the consistency of the regression coefficients.

(Module 1.3, LOS 1.h)

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Peter Pun, an enrolled candidate for the CFA Level II examination, has decided to perform a calendar test to examine whether there is any abnormal return associated with investments and disinvestments made in blue-chip stocks on particular days of the week. As a proxy for blue-chips, he has decided to use the S&P 500 Index. The analysis will involve the use of dummy variables and is based on the past 780 trading days. Here are selected findings of his study:

RSS            0.0039

SSE            0.9534

SST            0.9573

R-squared    0.004

SEE            0.035

Jessica Jones, CFA, a friend of Peter, overhears that he is interested in regression analysis and warns him that whenever heteroskedasticity is present in multiple regression, it could undermine the regression results. She mentions that one easy way to spot conditional heteroskedasticity is through a scatter plot, but she adds that there is a more formal test. Unfortunately, she can't quite remember its name. Jessica believes that heteroskedasticity

can be rectified using White-corrected standard errors. Her son Jonathan who has also taken part in the discussion, hears this comment and argues that White corrections would typically reduce the number of Type I errors in financial data.

### Question #115 - 118 of 139

Question ID: 1472057

What is *most likely* represented by the Y intercept of the regression?

- A) The drift of a random walk. 
- B) The intercept is not a driver of returns, only the independent variables. 
- C) The return on a particular trading day. 

#### Explanation

The omitted variable is represented by the intercept. So, if we have four variables to represent Monday through Thursday, the intercept would represent returns on Friday. Remember when we want to distinguish between "n" classes we always use one less dummy variable the number of classes ( $n - 1$ ).

(Module 1.4, LOS 1.I)

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### Question #116 - 118 of 139

Question ID: 1479957

What can be said of the overall explanatory power of the model at the 5% significance?

The coefficient of determination for the above regression is significantly higher

- A) than the standard error of the estimate, and therefore there is value to calendar trading. 
- B) There is no value to calendar trading. 
- C) There is value to calendar trading. 

#### Explanation

This question calls for a computation of the  $F$ -stat for all independent variables jointly.  $F = (0.0039 / 4) / (0.9534 / (780 - 4 - 1)) = 0.79$ . The critical  $F$  is somewhere between 2.37 and 2.45 so we fail to reject the null that the coefficient are equal to zero.

(Module 1.2, LOS 1.e)

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### Question #117 - 118 of 139

Question ID: 1472059

The test mentioned by Jessica is known as the:

- A) Breusch-Pagan, which is a two-tailed test. 
- B) Durbin-Watson, which is a two-tailed test. 
- C) Breusch-Pagan, which is a one-tailed test. 

#### Explanation

The Breusch-Pagan is used to detect conditional heteroskedasticity and it is a one-tailed test. This is because we are only concerned about large values in the residuals coefficient of determination.

(Module 1.3, LOS 1.h)

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### Question #118 - 118 of 139

Question ID: 1479958

Are Jessica and her son Jonathan correct in terms of the method used to correct for heteroskedasticity and the likely effects?

- A) Neither is correct. 
- B) Both are correct. 
- C) One is correct. 

#### Explanation

Jessica is correct. White-corrected standard errors are also known as robust standard errors. Jonathan is correct because for financial data, generally, White-corrected errors are higher than the biased errors leading to lower computed t-statistics and, therefore, less frequent rejection of the null hypothesis (remember incorrectly rejecting a true null is Type I error).

(Module 1.3, LOS 1.h)

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### Question #119 of 139

Question ID: 1479922

Which of the following is a potential remedy for multicollinearity?

- A) Take first differences of the dependent variable. 
- B) Add dummy variables to the regression. 
- C) Omit one or more of the collinear variables. 

#### Explanation

The first differencing is not a remedy for the collinearity, nor is the inclusion of dummy variables. The best potential remedy is to attempt to eliminate highly correlated variables.

(Module 1.3, LOS 1.i)

### Question #120 of 139

Question ID: 1472009

Which of the following statements regarding heteroskedasticity is *least* accurate?

- A) When not related to independent variables, heteroskedasticity does not pose any major problems with the regression. 
- B) Heteroskedasticity only occurs in cross-sectional regressions. 
- C) Conditional heteroskedasticity can be detected using the Breusch-Pagan chi-square statistic. 

#### Explanation

If there are shifting regimes in a time-series (e.g., change in regulation, economic environment), it is possible to have heteroskedasticity in a time-series. Unconditional heteroskedasticity occurs when the heteroskedasticity is not related to the level of the independent variables. Unconditional heteroskedasticity causes no major problems with the regression. Breusch-Pagan statistic has a chi-square distribution and can be used to detect conditional heteroskedasticity.

(Module 1.3, LOS 1.h)

Binod Salve, CFA, is investigating the application of the Fama-French three-factor model (Model 1) for the Indian stock market for the period 2001–2011 (120 months). Using the dependent variable as annualized return (%), the results of the analysis are shown in **Indian Equities—Fama-French Model**

#### Indian Equities—Fama-French Model

| Factor    | Coefficient | P-Value | VIF |
|-----------|-------------|---------|-----|
| Intercept | 1.22        | <0.001  |     |
| SMB       | 0.23        | <0.001  | 3   |
| HML       | 0.34        | 0.003   | 3   |
| Rm-Rf     | 0.88        | <0.001  | 2   |
| R-squared |             | 0.36    |     |

|            |       |
|------------|-------|
| SSE        | 38.00 |
| BG (lag 1) | 2.11  |
| BG (lag 2) | 1.67  |

**Partial F-Table (5% Level of Significance)**

| Degrees of Freedom Denominator | Degrees of Freedom Numerator |      |      |
|--------------------------------|------------------------------|------|------|
|                                | 1                            | 2    | 3    |
| 112                            | 3.93                         | 3.08 | 2.69 |
| 113                            | 3.93                         | 3.08 | 2.68 |
| 114                            | 3.92                         | 3.08 | 2.68 |
| 115                            | 3.92                         | 3.08 | 2.68 |
| 116                            | 3.92                         | 3.07 | 2.68 |
| 117                            | 3.92                         | 3.07 | 2.68 |

**Partial Chi-Square Table (5% Level of Significance)**

| Degrees of Freedom | Critical Value |
|--------------------|----------------|
| 1                  | 3.84           |
| 2                  | 5.99           |
| 3                  | 7.81           |
| 4                  | 9.49           |
| 5                  | 11.07          |
| 6                  | 12.59          |

### Question #121 - 124 of 139

Question ID: 1501597

Salve runs a regression using the squared residuals from the model using the original dependent variables. The coefficient of determination of this model is 6%. Which of the following is the *most appropriate* conclusion at a 5% level of significance?

- A) Because the test statistic of 7.20 is higher than the critical value of 3.84, we reject the null hypothesis of no conditional heteroskedasticity in residuals.



- B) Because the test statistic of 7.20 is lower than the critical value of 7.81, we fail to reject the null hypothesis of no conditional heteroskedasticity in residuals. 
- C) Because the test statistic of 3.60 is lower than the critical value of 3.84, we reject the null hypothesis of no conditional heteroskedasticity in residuals. 

### Explanation

The chi-square test statistic =  $n \times R^2 = 120 \times 0.06 = 7.20$ .

The one-tailed critical value for a chi-square distribution with  $k = 3$  degrees of freedom and  $\alpha$  of 5% is 7.81. Therefore, we should not reject the null hypothesis and conclude that we don't have a problem with conditional heteroskedasticity.

(Module 1.3, LOS 1.h)

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### Question #122 - 124 of 139

Question ID: 1501598

Which of the following misspecifications is *most likely* to cause serial correlation in residuals?

- A) Data improperly pooled. 
- B) Improper variable scaling. 
- C) Improper variable form. 

### Explanation

Out of the four forms of model misspecifications, serial correlation in residuals may be caused by omission of important variables (not an answer choice) and by improper data pooling.

(Module 1.3, LOS 1.g)

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### Question #123 - 124 of 139

Question ID: 1501599

Should Salve be concerned about residual serial correlation?

- A) Yes, for one lag only. 
- B) Yes, for two lags only. 
- C) No. 

### Explanation

The BG test statistic has an  $F$ -distribution with  $p$  and  $n - p - k - 1$  degrees of freedom, where  $p$  = the number of lags tested. Given  $n = 120$  and  $k = 3$ , critical  $F$ -values (5% level of significance) are 3.92 ( $p = 1$ ) and 3.08 ( $p = 2$ ). BG stats in **Indian Equities—Fama-French Model** are lower than the critical  $F$ -values; therefore, serial correlation does not seem to be a problem for both lags.

(Module 1.3, LOS 1.i)

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### Question #124 - 124 of 139

Question ID: 1501600

Should Salve be concerned about residual multicollinearity?

- A) Yes, and Salve should exclude either variable SMB or HML from the model. 
- B) Yes, and Salve should exclude variable Rm-Rf from the model. 
- C) No. 

### Explanation

Multicollinearity is detected using the variance inflation factor (VIF). VIF values greater than 5 (i.e.,  $R^2 > 80\%$ ) warrant further investigation, while values above 10 (i.e.,  $R^2 > 90\%$ ) indicate severe multicollinearity. None of the variables have  $VIF > 5$ .

(Module 1.3, LOS 1.j)

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### Question #125 of 139

Question ID: 1472067

An analyst runs a regression of portfolio returns on three independent variables. These independent variables are price-to-sales (P/S), price-to-cash flow (P/CF), and price-to-book (P/B). The analyst discovers that the p-values for each independent variable are relatively high. However, the F-test has a very small p-value. The analyst is puzzled and tries to figure out how the F-test can be statistically significant when the individual independent variables are not significant. What violation of regression analysis has occurred?

- A) multicollinearity. 
- B) conditional heteroskedasticity. 
- C) serial correlation. 



### Explanation

An indication of multicollinearity is when the independent variables individually are not statistically significant but the F-test suggests that the variables as a whole do an excellent job of explaining the variation in the dependent variable.

(Module 1.3, LOS 1.j)

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### Question #126 of 139

Question ID: 1471946

Which of the following statements regarding the  $R^2$  is *least* accurate?

- A) The adjusted- $R^2$  is greater than the  $R^2$  in multiple regression. 
- B) It is possible for the adjusted- $R^2$  to decline as more variables are added to the multiple regression. 
- C) The adjusted- $R^2$  not appropriate to use in simple regression. 

### Explanation

The adjusted- $R^2$  will always be less than  $R^2$  in multiple regression.

(Module 1.2, LOS 1.d)

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### Question #127 of 139

Question ID: 1472007

An analyst is trying to estimate the beta for a fund. The analyst estimates a regression equation in which the fund returns are the dependent variable and the Wilshire 5000 is the independent variable, using monthly data over the past five years. The analyst finds that the correlation between the square of the residuals of the regression and the Wilshire 5000 is 0.2. Which of the following is *most* accurate, assuming a 0.05 level of significance? There is:

- A) no evidence that there is conditional heteroskedasticity or serial correlation in the regression equation. 
- B) evidence of conditional heteroskedasticity but not serial correlation in the regression equation. 
- C) evidence of serial correlation but not conditional heteroskedasticity in the regression equation. 

### Explanation

The test for conditional heteroskedasticity involves regressing the square of the residuals on the independent variables of the regression and creating a test statistic that is  $n \times R^2$ , where  $n$  is the number of observations and  $R^2$  is from the squared-residual regression. The test statistic is distributed with a chi-squared distribution with the number of degrees of freedom equal to the number of independent variables. For a single variable, the  $R^2$  will be equal to the square of the correlation; so in this case, the test statistic is  $60 \times 0.2^2 = 2.4$ , which is less than the chi-squared value (with one degree of freedom) of 3.84 for a p-value of 0.05. There is no indication about serial correlation.

(Module 1.3, LOS 1.h)

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### Question #128 of 139

Question ID: 1479902

When pooling the samples over multiple economic environments in a multiple regression model, which of the following errors is *most likely* to occur?

- A) Heteroskedasticity.
- B) Model misspecification.
- C) Multicollinearity.



#### Explanation

When data are improperly pooled over multiple economic environments in a multiple regression analysis, the model would be misspecified.

(Module 1.3, LOS 1.g)

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### Question #129 of 139

Question ID: 1586002

Which of the following statements regarding the  $R^2$  is *least* accurate?

- A)  $R^2$  is the coefficient of determination of the regression.
- B) The  $R^2$  of a regression will be greater than or equal to the adjusted- $R^2$  for the same regression.
- C) The  $R^2$  is the ratio of the unexplained variation to the explained variation of the dependent variable.



#### Explanation

The  $R^2$  is the ratio of the explained variation to the total variation.

(Module 1.2, LOS 1.d)

Manuel Mercado, CFA has performed the following two regressions on sales data for a given industry. He wants to forecast sales for each quarter of the upcoming year.

| Model ONE             |          |
|-----------------------|----------|
| Regression Statistics |          |
| Multiple R            | 0.941828 |
| $R_2$                 | 0.887039 |
| Adjusted $R_2$        | 0.863258 |
| Standard Error        | 2.543272 |
| Observations          | 24       |

Durbin-Watson test statistic = 0.7856

| ANOVA      |    |           |          |          |                |
|------------|----|-----------|----------|----------|----------------|
|            | df | SS        | MS       | F        | Significance F |
| Regression | 4  | 965.0619  | 241.2655 | 37.30006 | 9.49E-09       |
| Residual   | 19 | 122.8964  | 6.4682   |          |                |
| Total      | 23 | 1087.9583 |          |          |                |

|           | Coefficients | Standard Error | t-Statistic |
|-----------|--------------|----------------|-------------|
| Intercept | 31.40833     | 1.4866         | 21.12763    |
| Q1        | -3.77798     | 1.485952       | -2.54246    |
| Q2        | -2.46310     | 1.476204       | -1.66853    |
| Q3        | -0.14821     | 1.470324       | -0.10080    |
| TREND     | 0.851786     | 0.075335       | 11.20848    |

| Model TWO             |          |
|-----------------------|----------|
| Regression Statistics |          |
| Multiple R            | 0.941796 |

|                         |          |
|-------------------------|----------|
| R <sup>2</sup>          | 0.886979 |
| Adjusted R <sup>2</sup> | 0.870026 |
| Standard Error          | 2.479538 |
| Observations            | 24       |

Durbin-Watson test statistic = 0.7860

|            | df | SS        | MS       | F       | Significance F |
|------------|----|-----------|----------|---------|----------------|
| Regression | 3  | 964.9962  | 321.6654 | 52.3194 | 1.19E-09       |
| Residual   | 20 | 122.9622  | 6.14811  |         |                |
| Total      | 23 | 1087.9584 |          |         |                |

|           | Coefficients | Standard Error | t-Statistic |
|-----------|--------------|----------------|-------------|
| Intercept | 31.32888     | 1.228865       | 25.49416    |
| Q1        | -3.70288     | 1.253493       | -2.95405    |
| Q2        | -2.38839     | 1.244727       | -1.91881    |
| TREND     | 0.85218      | 0.073991       | 11.51732    |

The dependent variable is the level of sales for each quarter, in \$ millions, which began with the first quarter of the first year. Q1, Q2, and Q3 are seasonal dummy variables representing each quarter of the year. For the first four observations the dummy variables are as follows: Q1:(1,0,0,0), Q2:(0,1,0,0), Q3:(0,0,1,0). The TREND is a series that begins with one and increases by one each period to end with 24. For all tests, Mercado will use a 5% level of significance. Tests of coefficients will be two-tailed, and all others are one-tailed.

### Question #130 - 134 of 139

Question ID: 1471938

Which model would be a better choice for making a forecast?

- A) Model TWO because it has a higher adjusted R<sup>2</sup>.
- B) Model TWO because serial correlation is not a problem.
- C) Model ONE because it has a higher R<sup>2</sup>.



**Explanation**

Model TWO has a higher adjusted  $R^2$  and thus would produce the more reliable estimates. As is always the case when a variable is removed,  $R^2$  for Model TWO is lower. The increase in adjusted  $R^2$  indicates that the removed variable, Q3, has very little explanatory power, and removing it should improve the accuracy of the estimates. With respect to the references to autocorrelation, we can compare the Durbin-Watson statistics to the critical values on a Durbin-Watson table. Since the critical DW statistics for Model ONE and TWO respectively are 1.01 ( $>0.7856$ ) and 1.10 ( $>0.7860$ ), serial correlation is a problem for both equations.

(Module 1.2, LOS 1.d)

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### Question #131 - 134 of 139

Question ID: 1479951

Using Model ONE, what is the sales forecast for the second quarter of the next year?

- A) \$51.09 million.
- B) \$46.31 million.
- C) \$56.02 million.



#### Explanation

The estimate for the second quarter of the following year would be (in millions):

$$31.4083 + (-2.4631) + (24 + 2) \times 0.851786 = 51.091666.$$

(Module 1.2, LOS 1.f)

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### Question #132 - 134 of 139

Question ID: 1479952

Which model misspecification is *most likely* to cause multicollinearity?

- A) Inappropriate variable scaling.
- B) Inappropriate variable form.
- C) Omission of important variable(s).



#### Explanation

Inappropriate variable scaling may lead to multicollinearity or heteroskedasticity in residuals. Omission of important variable may lead to biased and inconsistent regression parameters and also heteroskedasticity/serial correlation in residuals. Inappropriate variable form can lead to heteroskedasticity in residuals.

(Module 1.3, LOS 1.g)

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**Question #133 - 134 of 139**

Question ID: 1479953

If it is determined that conditional heteroskedasticity is present in model one, which of the following inferences are *most* accurate?

- A) Regression coefficients will be unbiased but standard errors will be biased. 
- B) Regression coefficients will be biased but standard errors will be unbiased. 
- C) Both the regression coefficients and the standard errors will be biased. 

**Explanation**

Presence of conditional heteroskedasticity will not affect the consistency of regression coefficients but will bias the standard errors leading to incorrect application of t-tests for statistical significance of regression parameters.

(Module 1.3, LOS 1.h)

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**Question #134 - 134 of 139**

Question ID: 1479955

If Mercado determines that Model TWO is the appropriate specification, then he is essentially saying that for each year, value of sales from quarter three to four is expected to:

- A) grow by more than \$1,000,000. 
- B) remain approximately the same. 
- C) grow, but by less than \$1,000,000. 

**Explanation**

The specification of Model TWO essentially assumes there is no difference attributed to the change of the season from the third to fourth quarter. However, the time trend is significant. The trend effect for moving from one season to the next is the coefficient on TREND times \$1,000,000 which is \$852,182 for Equation TWO.

(Module 1.1, LOS 1.b)

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In preparing an analysis of Treefell Company, Jack Lumber is asked to look at the company's sales in relation to broad-based economic indicators. Lumber's analysis indicates that Treefell's monthly sales are related to changes in housing starts (H) and changes in the mortgage interest rate (M). The analysis covers the past 10 years for these variables. The regression equation is:

$$S = 1.76 + 0.23H - 0.08M$$

Number of observations: 123

Unadjusted  $R^2$ : 0.77

$F$ -statistic: 9.80

Durbin-Watson statistic: 0.50

$p$ -value of Housing Starts: 0.017

$t$ -stat of Mortgage Rates: -2.6

### Variable Descriptions

S = Treefell Sales (in thousands)

H = housing starts (in thousands)

M = mortgage interest rate (in percent)

### November 20X6 Actual Data

Treefell's monthly sales: \$55,000

Housing starts: 150,000

Mortgage interest rate (%): 7.5

### Partial Chi-Square Table (5% Level of Significance)

| Degrees of Freedom | Critical Value |
|--------------------|----------------|
| 1                  | 3.84           |
| 2                  | 5.99           |
| 3                  | 7.81           |
| 4                  | 9.49           |

5 11.07

6 12.59

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### Question #135 - 139 of 139

Question ID: 1472050

Using the regression model developed, the *closest* prediction of sales for December 20X6 is:

A) \$55,000.



B) \$44,000.



C) \$36,000.



#### Explanation

$$1.76 + 0.23 \times (150) - 0.08 \times (7.5) = 35.66.$$

(Module 1.2, LOS 1.f)

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### Question #136 - 139 of 139

Question ID: 1489312

Will Jack conclude that the housing starts coefficient is statistically different from zero and how will he interpret it at the 5% significance level?

A) Different from zero; sales will rise by \$100 for every 23 house starts.



B) Different from zero; sales will rise by \$23 for every 100 house starts.



C) Not different from zero; sales will rise by \$0 for every 100 house starts.



#### Explanation

A  $p$ -value (0.017) below significance (0.05) indicates a variable that is statistically different from zero. The coefficient of 0.23 indicates that sales will rise by \$23 for every 100 house starts. Remember the rule  $p\text{-value} < \text{significance}$ , then reject null.

(Module 1.1, LOS 1.b)

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### Question #137 - 139 of 139

Question ID: 1479916

In this multiple regression, the  $F$ -statistic indicates the:



- deviation of the estimated values from the actual values of the dependent variable.
- A) 
- B) degree of correlation between the independent variables. 
- C) the joint significance of the independent variables. 

#### Explanation

The  $F$ -statistic is for the general linear  $F$ -test to test the null hypothesis that slope coefficients on all variables are equal to zero.

(Module 1.2, LOS 1.e)

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### Question #138 - 139 of 139

Question ID: 1472054

The regression statistics indicate that for the period under study, the independent variables (housing starts, mortgage interest rate) together explain approximately what percentage of the variation in the dependent variable (sales)?

- A) 67.00. 
- B) 9.80. 
- C) 77.00. 

#### Explanation

The question is asking for the coefficient of determination.

(Module 1.2, LOS 1.d)

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### Question #139 - 139 of 139

Question ID: 1484386

For this question only, assume that the regression of squared residuals on the independent variables has  $R^2 = 11\%$ . At a 5% level of significance, which of the following conclusions is *most* accurate?

- A) With a test statistic of 13.53, we can conclude the presence of conditional heteroskedasticity. 
- B) With a test statistic of 0.22, we cannot reject the null hypothesis of no conditional heteroskedasticity. 

C) Because the critical value is 3.84, we reject the null hypothesis of no conditional heteroskedasticity.



**Explanation**

Chi-square =  $n \times R^2 = 123 \times 0.11 = 13.53$ . Critical Chi-square (degree of freedom =  $k = 2$ ) = 5.99. Because the test statistic exceeds the critical value, we reject the null hypothesis (of no conditional heteroskedasticity).

(Module 1.3, LOS 1.h)